Moving Government, Moving People? The Effects of Public-Sector Relocation

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Abstract. Public-sector relocations are a popular tool for balanced regional growth, given their low cost and expected impact on private sector growth. This paper examines the effects of such relocations using a quantitative spatial general equilibrium model. The model is quantified for South Korea and used to examine the general equilibrium effects of the Innovation City Project — a public-sector relocation that moved over 42,000 workers from the Seoul Metropolitan Area to ten designated sites across the country between 2011 and 2018. The project was accompanied by large-scale residential construction to house the incoming workers and attract additional private-sector employees. The model consists of several regions with limited land supply and three sectors: a public sector, a private tradable sector, and a private non-tradable sector. Households can commute to work and therefore choose their utility-maximizing residence-workplace location, accounting for goods and land prices at their residence, wages at their workplace, the disutility of commuting, and idiosyncratic amenities. The results show that, on average, each relocated public-sector worker increases employment in the non-tradable sector by 0.62 workers, while crowding out 0.24 workers in the tradable sector at the destination. Impacts of the policy are highly localized to the sending and receiving locations, with limited spillovers to surrounding regions.

1 Introduction

Large economic imbalances between Seoul, South Korea's capital, and the rest of the country are a persistent feature of South Korean policy debates. The issue has been a major component of every National Territorial Plan, the South Korean government's framework for national spatial development, since the 1980s (KRIHS 2022c, 2022d, 2022b, 2022a). Despite continuous policy interventions, concentration remains high: close to 50 % of the population lives in the Seoul Metropolitan Area, raising concerns about lagging peripheral regions and overcrowding in the capital. One solution favored by policymakers in other countries facing similar challenges is the relocation of public-sector workers. Such policies aim to revitalize left-behind regions at lower fiscal cost than other place-based policies.[1] Public-sector relocations have been implemented at the national level in countries such as the UK (Lyons 2004) and Denmark (Finansministeriet 2015), or at the sub-national level in Bavaria, Germany (StMFH 2014). Acting as a local engine for new growth, relocated public-sector jobs are expected to attract private sector activity by fostering local demand and improving local productivity.

This paper examines the employment effects of the Innovation City Project, South Korea's implementation of a public-sector relocation policy. Agreed upon in 2007, the project began relocations in 2011. It relocated over 42,000 workers from 135 government agencies, state-owned enterprises, and research institutes in the Seoul Metropolitan Area to ten districts, so-called "Innovation Cities", outside the capital. The main goal of the policy was to revitalize areas outside the Seoul Metropolitan Area that had fallen behind in economic growth. The relocated agencies and their workers were intended to serve as anchors for new growth centers, attracting additional private sector activity. Each Innovation City received between 752 and 7,043 public-sector jobs, often accounting for significant shares of local em-

1. Relocations are often presented as budget neutral, because the cost of the move can be offset by savings from vacating expensive capital premises (Lyons 2004; PAPPC 2008).

ployment. In addition to the worker relocations, the Innovation Cities benefited from government-backed residential development and public investment in amenity facilities. Over 100,000 residential units to house relocated workers and incoming private sector employees were constructed under the policy.

The paper provides reduced-form evidence on private-sector employment growth in manufacturing and services, which correspond closely to the tradable and non-tradable sectors in the structural model. A structural estimation then examines the general-equilibrium effects on the South Korean economy as a whole. The structural estimates show net gains in private-sector employment, driven by growth in the non-tradable sector, which aligns with the reduced-form evidence for services, while partially offset by crowding out in the tradable sector, the model counterpart to manufacturing.

A structural approach allows examination of not just the partial equilibrium effect on the designated cities, as measured in reduced-form studies, but also the general equilibrium effects on the broader economy. While employment gains at the Innovation Cities might imply policy success, the origin of those gains matters. The policy evaluation differs according to whether those gains stem from workers relocating from other peripheral regions or from the Seoul Metropolitan Area. The former would exacerbate regional inequalities, while the latter would reduce them. Given South Korea's excessive concentration of economic activity in the Seoul Metropolitan Area, these distributional questions are particularly important.

This paper constructs a spatial model with multiple regions, limited land supply, and bilateral flows of goods and workers, building on Monte, Redding, and Rossi-Hansberg (2018). The model features three sectors: a publicsector in which the central government assigns public servants to their workplaces, a tradable private sector, and a non-tradable private sector. Workers choose their optimal residence-workplace combination, consuming housing, tradable goods, and locally produced non-tradable goods representing consumer services at their place of residence. The

paper uses the model to simulate the Innovation City Project, incorporating both the relocation of public-sector workers and the accompanying housing construction in the Innovation Cities. The simulation draws on data covering South Korean wages, house prices, bilateral commuting flows, public-sector employment shares, and the destination and employment size of each relocated agency.

The counterfactual results show that each relocated public-sector worker attracts 0.62 additional private sector workers in the non-tradable sector while crowding out 0.24 workers in the tradable sector. The policy's impact on residential location remains highly localized: 90 % of the net outflow of residents from the Seoul Metropolitan Area relocates to the Innovation Cities. Net employment changes occur almost entirely in the Seoul Metropolitan Area and the Innovation Cities. Outside these locations tradable employment shifts to counties farther from the Innovation Cities, but this effect is offset by gains in non-tradable employment. Net employment in counties outside the Innovation Cities therefore remains essentially unchanged.

This paper contributes to the place-based policy literature along three dimensions: the tradeoff between local productivity gains and cost pressures arising from public-sector relocations, the job creation mechanism, and the distinctive general equilibrium effects of relocations. Like evaluations of Special Economic Zones in China (Wang 2013; Lu, Wang, and Zhu 2019) and Urban Empowerment Zones in the United States (Busso, Gregory, and Kline 2013), this paper examines the tradeoff between productivity gains for local firms from agglomeration economies and cost pressures from higher local demand for land and labor. The analysis traces how this tradeoff shapes local economic activity. The spatial framework also enables examination of the factors driving variation in policy effectiveness across the designated cities, following Briant, Lafourcade, and Schmutz (2015).

Public-sector relocations differ from conventional place-based policies in the mechanism through which they generate local employment. Conventional incentive-based interventions, including special economic zones, employment subsidies, invest-

ment subsidies, and infrastructure provision, induce private actors to expand local economic activity (Neumark and Simpson 2015). The effectiveness of the initial supply stimulus depends on private sector responses, which can prove limited, as with Opportunity Zones in the United States (Freedman, Khanna, and Neumark 2023) and Enterprise Zones in France (Gobillon, Magnac, and Selod 2012). Public-sector relocations bypass this contingency: rather than altering the incentives facing private actors, the government directly assigns workers to the target location. The size of the initial employment shock is therefore observed and contractually fixed, rather than the result of private responses. This aids identification in the reduced-form analysis and calibration of the structural model, since policy take-up need not be estimated or modeled. Because the trigger is exogenous to private behavior, the downstream effects on non-tradable and tradable private employment can be attributed to the policy more cleanly than with incentive-based alternatives. Incentive-based policies such as employment subsidies create a confounding problem: some employment gains are direct responses to the incentive, while others are second-order effects from higher local productivity or demand. The two are difficult to disentangle. With a public-sector relocation, the initial employment shock is directly observable: administrative records capture the number of relocated workers, so the remaining employment changes are attributable to private sector responses.

Conventional place-based policies are typically financed from a central budget, giving evaluations a clearly defined target location but no identifiable sending location. Consequently, the literature tends to focus on local treatment effects and the efficiency of public spending (Blouri and Von Ehrlich 2020; Becker, Egger, and Von Ehrlich 2010). Infrastructure interventions such as the Tennessee Valley Authority illustrate this setting: they affect the target region and its surroundings, but they do not directly transfer economic activity from a specific origin region (Kline and Moretti 2014). In such cases, disentangling the regional incidence of the tax burden is not essential for evaluating the policy's spatial effects. Public-sector relocations

are different because they involve both a target and an origin. In the South Korean context, Seoul serves as the sending location, allowing the evaluation to examine how the policy affects both the origin region and the receiving regions.

The paper also contributes to the literature on local employment multipliers, particularly those of public-sector workers. Following Moretti (2010), this literature (e.g., Moretti and Thulin 2013; Osman and Kemeny 2022; De Blasio and Menon 2011; Bartik and Sotherland 2019) examines the additional employment created by shocks to local demand or local productivity. For public-sector multipliers specifically, estimated effect sizes vary: Jensen, Schmidt, and Sørensen (2025) find negligible employment effects for the Danish relocations, while Jofre-Monseny, Silva, and Vázquez-Grenno (2020) find multipliers of up to 1.8 for a public-sector expansion in Spain. This paper provides reduced-form evidence on the local employment multiplier of the Innovation City Project, as do Lee, Ko, and Kim (2024). Contrary to Lee, Ko, and Kim (2024), however, this paper limits the analysis to relocations into the Innovation Cities, excluding isolated relocations to other districts that occurred under the same policy. The public-sector relocation literature generally finds that employment gains are concentrated in the service sector, with weak or negative effects on manufacturing, as in Becker, Heblich, and Sturm (2021) for Germany and Faggio and Overman (2014) for the UK. The structural model provides intuition on the factors driving the extent of this manufacturing displacement.

Finally, this paper contributes to the literature on the Innovation City Project in South Korea. Alongside Lee, Ko, and Kim (2024), Kang, Lee, and Kim (2024) examine migration effects, focusing on the in- and out-migration of young residents. Seo and Do (2023) study the impact of Innovation City designation on employment and productivity, finding marginal private sector employment gains and modest productivity effects for established manufacturing firms in the years following the relocations. This paper complements these reduced-form findings by providing a structural explanation for the limited productivity gains and the net migration flows.

The model builds on Monte, Redding, and Rossi-Hansberg (2018), extending it to incorporate a public sector. This paper therefore contributes to the structural literature on the role of the public sector in spatial equilibrium. While much of this literature focuses on fiscal dimensions such as regional transfers (Henkel, Seidel, and Suedekum 2021) and taxes (Fajgelbaum et al. 2019), or on the market power of the public sector (Guillouzouic, Henry, and Monras 2024), this paper examines public-sector employment and its spatial reallocation.

The work most closely related to this paper comes from Freitas (2025) and Jofre-Monseny, Silva, and Vázquez-Grenno (2020). Jofre-Monseny, Silva, and Vázquez-Grenno (2020) construct a search and matching model for a local labor market with public and private sectors. The paper focuses on local wage responses and unemployment effects arising from increases in public-sector hiring. This paper complements that work by introducing a spatial component, examining general equilibrium effects for the country as a whole and allowing workers to commute between districts. Freitas (2025) similarly examines public-sector relocation as a place-based policy, focusing on the Homeland Strategy in Bavaria, Germany. Aside from differences in national context and scale of the relocation, the two papers differ in their assumptions and modeling approach. Most notably, this paper focuses on the differential effects on the tradable and non-tradable private sectors, while Freitas (2025) emphasizes productivity effects. Freitas (2025) models a single private sector but includes productivity spillovers between regions and complementarities between public and private workers that jointly affect local productivity. Since this paper examines effects at a more aggregated spatial scale, it omits productivity spillovers between regions and models productivity through agglomeration economies alone, following Combes et al. (2012). Productivity here depends only on a location-specific exogenous component and the total number of workers, whereas in Freitas (2025) it additionally depends on the composition of public and private workers.

Section 2 provides the institutional background of the Innovation City Project.

Section 3 describes the data. Section 4 presents reduced-form evidence on the Innovation Cities and their post-designation development. Section 5 introduces the theoretical model, including its equilibrium conditions. Section 6 describes model calibration and parameter recovery. Section 7 presents the counterfactual simulations and results, along with robustness checks. The final section concludes.

2 Institutional Background

South Korea has one of the highest concentrations of population and economic activity in its capital region among OECD countries: close to 50 % of the population resides in the Seoul Metropolitan Area (see Table E.1 in the Appendix). This concentration represents a substantial deviation from Zipf's law (Veneri 2013; Soo 2005): the Seoul Metropolitan Area substantially exceeds its Zipf-predicted size, while secondary and tertiary cities fall well below theirs.

The uneven distribution of economic activity stems from South Korean economic policy in the aftermath of the Korean War. While these policies led to rapid economic development, growth concentrated heavily in specific regions, particularly the Seoul Metropolitan Area. The spatial pattern reflected a deliberate policy of directing scarce capital toward specific regions. Working jointly with large private conglomerates, state and private investment focused on specific manufacturing industries in the Seoul Metropolitan Area and the southeastern region.[2] In the early postwar decades, both regions developed rapidly, with the southeastern region concentrating much of South Korean heavy industry, including steel production, chemicals, and shipbuilding. By the late 20th century, however, growth increasingly concentrated in the Seoul Metropolitan Area, while activity slowed in the southeastern region. The agglomeration of company headquarters and a large pool of human

2. The Seoul Metropolitan Area includes Seoul itself, the neighboring municipality and port city of Incheon, and the surrounding Gyeonggi province, while the southeastern region contains Busan, the second-largest city in South Korea, the municipality of Ulsan, and the surrounding cities in the provinces of Gyeongsangnam and Gyeongsangbuk.

capital made the Seoul Metropolitan Area better suited for the shift toward high-tech manufacturing and knowledge-based services. These industries increasingly drove economic development over this period (OECD 2012). These developments produced a heavy concentration of population and economic, cultural, and political power in the Seoul Metropolitan Area (see Table E.2 in the Appendix). By the early 21st century, 90% of companies and 85% of public institutions were headquartered there.

While this concentration of activity was the result of a successful development policy, it also created several challenges for the country. The concentration of growth in the Seoul Metropolitan Area limited development in other regions. The resulting stagnation drove a sustained outflow of young and educated workers from peripheral regions to the Seoul Metropolitan Area. This dynamic reinforced regional imbalances, fostered discontent in left-behind regions, and caused overcrowding in the Seoul Metropolitan Area. To alleviate these challenges, South Korea introduced several policies aimed at reducing concentration in the Seoul Metropolitan Area and promoting more balanced regional development (Kim et al. 2006). South Korea therefore provides a valuable research environment for examining place-based policies. The regional policies introduced at the turn of the 21st century have a clear equity focus, aiming at supporting left-behind regions. However, for any of these policies to be successful they must overcome the significant benefits of locating in the Seoul Metropolitan Area to foster development elsewhere.

2.1 Innovation City Project

The Innovation City Project, implemented between 2011 and 2018, was one of the most explicit responses to the rising concentration in the Seoul Metropolitan Area introduced by the South Korean government. The project aimed to redirect economic activity to designated Innovation Cities by relocating public-sector institutions from the Seoul Metropolitan Area. It formed part of a broader set of "New

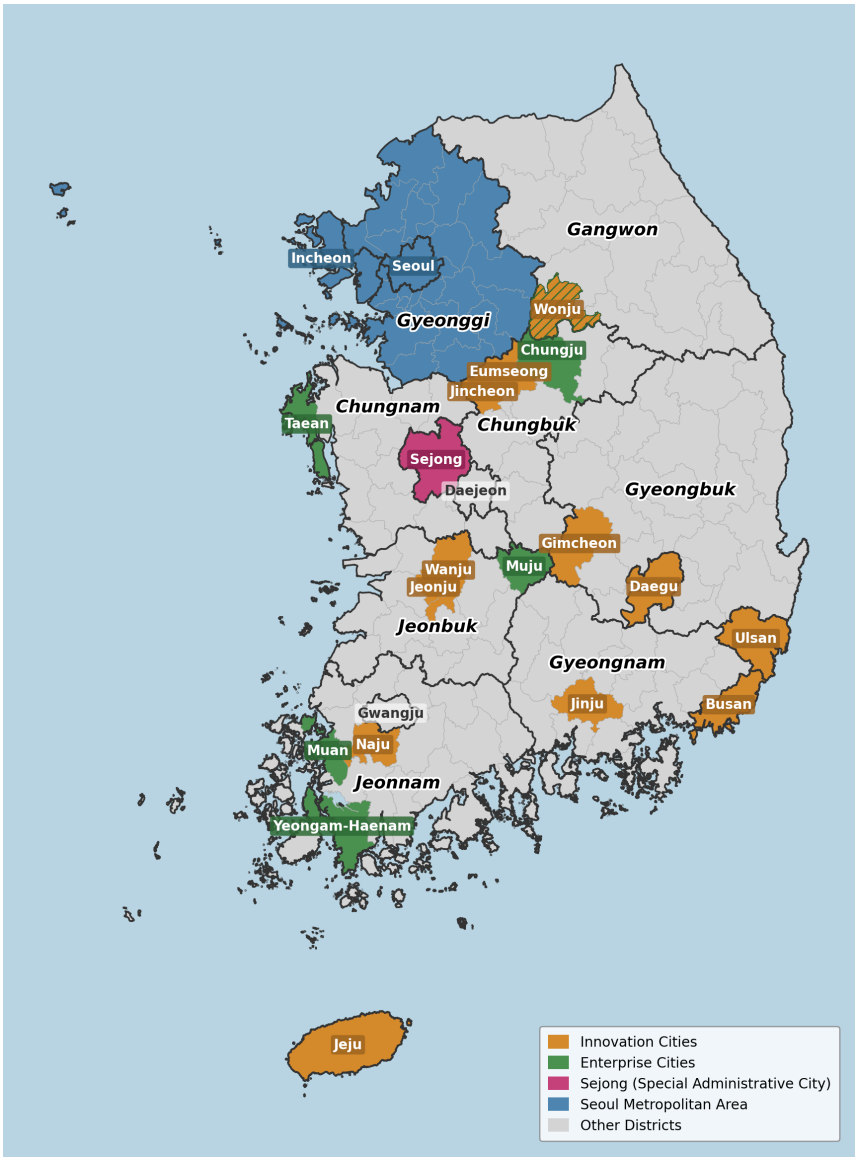
Town" projects (see Figure 1), government-planned urban districts that have been part of South Korean regional policy since the 1960s (Lee, You, and Kwon 2015). Discussions began as part of a broader balanced-growth agenda codified in the "Special Act on Balanced National Development" of 2003 (PAPPC 2008), with the policy process continuing until 2007. The relocated institutions were intended to attract private sector activity by establishing the Innovation Cities as viable alternatives to the Seoul Metropolitan Area. Designation occurred at the county level, with either provincial counties or neighborhoods within municipalities selected as sites.[3] Construction at most Innovation Cities began in 2007 under the "Innovation City Special Act". Staff relocations followed in 2011 and continued until 2018, with the bulk occurring between 2013 and 2015.[4]

The policy process divided into three stages: the selection of institutions for relocation; the distribution of those institutions among the provinces; and the selection of the Innovation City site within each province. Each stage typically involved deliberations among all major stakeholders, including central and provincial governments, representatives of the public institutions, and public-sector unions. Consistent with the broader pattern of South Korean place-based policies, the central government set the overall policy direction and provided the legal framework, while local governments were required to approve implementation but retained no final decision-making authority. Various committees developed implementation proposals, typically comprising representatives of the relevant parties. The selection of institutions for relocation, for example, was decided by the "Special Committee on the Relocation of Public Institutions to the Provinces". The committee drew on officials from the relevant central government ministries, representatives of the public institutions, and academics providing expert advice.

3. Throughout, *district* denotes the third administrative layer in South Korea, *county* the second, and *province* the first. For a more detailed description of the South Korean administrative system, see Appendix B.

4. A second phase of the program was launched in 2018. Unlike the first phase, it focused on developing Innovation Cities into regional innovation hubs rather than on further public-agency relocations or new housing construction (Yoon 2019).

Figure 1: Innovation City and Other New Town Locations



Notes: Map shows counties aggregated according to Appendix B with province borders. New Town projects in orange (Innovation Cities), green (Enterprise Cities) and purple (Sejong Administrative City). Seoul Metropolitan Area in blue.

Public institutions eligible for relocation included all agencies subordinate to the central government and state-owned corporations. The policy excluded institutions with direct connections to the Seoul Metropolitan Area (such as those managing local heritage sites) and those facing significant efficiency losses or excessive moving costs (such as Seoul National Hospital). Further exemptions applied to institutions central to Seoul's international economic standing (such as the Korea Development Bank) and to those with large private shareholders who could not be compelled to relocate (such as Korea First Bank). Of the 345 institutions located in the Seoul Metropolitan Area, 245 were considered for relocation, with 175 designated as suitable in 2005 (PAPPC 2008).

Also in 2005, the South Korean government decided to construct the Special Self-Governing City of Sejong, adding a further destination for public-sector relocations. Initially planned as the new national capital, with the entire government to be relocated to Sejong, the project was scaled back under legal and political pressure. Nonetheless, the central government relocated several designated institutions and several ministries to the new city. Of the 175 institutions designated for relocation, 40 were moved to Sejong, leaving 135 for the Innovation Cities. Research institutions made up a disproportionate share of the Sejong relocations, accounting for 17 of the 40 (PAPPC 2008).

The Innovation City Project designated one Innovation City per province to spread growth across the country. It excluded the three administrative units of the Seoul Metropolitan Area: Seoul, Incheon, and Gyeonggi-do. Chungnam province was also excluded, as it was designated to host the new administrative capital of Sejong. The project further excluded the municipality of Daejeon owing to its high level of public-sector employment. This reflected both an earlier relocation that had brought agencies such as the customs office and the national archives, and the presence of several prominent universities.[5] Finally, Jeonnam province and the

5. Both Daejeon and Chungnam received Innovation City designations in 2020; however, these designations were not accompanied by sizable public-sector relocations.

neighboring municipality of Gwangju established a joint Innovation City in Naju, which directly borders Gwangju, bringing the total number of Innovation Cities to ten.

With the institutions for relocation identified, deliberations over their provincial distribution began in the second half of 2005. The selection sought to balance two partially competing goals: placing institutions in locations suited to their operational efficiency and regional productivity, and directing resources toward weaker provinces on equity grounds. Fierce inter-provincial competition for high-value institutions further complicated the process. Large state-owned enterprises were especially coveted for the workers and tax revenues they would bring; the Korea Electric Power Corporation (KEPCO), the national energy utility, attracted the greatest competition. Eight of the ten targeted provinces ranked KEPCO as their first preference. The assignment ultimately prioritized the balanced growth objective, giving preferential treatment to economically weaker provinces. KEPCO was accordingly assigned to Jeonnam, one of the poorer provinces in the southwest (PAPPC 2008).

With the provincial distribution settled, each province began selecting the site for its Innovation City. The project mandated a single Innovation City per province, concentrating all relocations and construction efforts in one location to create a dedicated innovation hub.[6] In municipalities, the process was straightforward, as the municipal government was the sole political actor. In some provinces, however, county officials competed fiercely over which county should receive the Innovation City designation. The same tension between balanced growth and development potential that had shaped the provincial allocation resurfaced at the site selection stage. The economically strongest counties argued for their own designation on grounds of development potential, while weaker counties invoked the balanced growth rationale. The latter warned that further concentrating investment in the province's most dy-

6. For a small number of special cases individual agencies were located outside the Innovation City, if the role of the agency made it necessary. For example the Korea Forest Aviation Headquarters were moved to a rural district near the actual Innovation City.

namic county would hollow out the rest of the province. Site selection deliberations ran from late 2005 to 2007 (PAPPC 2008). Unlike the provincial allocation, the Innovation City location criteria shown in Table E.3 favored economically stronger counties, which tended to be better connected to the South Korean transport network and showed greater potential for growth. Figure 1 maps the final distribution of Innovation Cities alongside concurrent New Town construction projects. The Innovation Cities in Jeonbuk and Chungbuk straddle the border between two counties, with institutions on both sides.

Table 1 confirms the emphasis on development potential in the site selection. Almost all designated counties rank in the upper half of their respective province by wages, rents, and employment. The joint Innovation City arrangement between Jeonnam and Gwangju constrained site selection within Jeonnam to counties neighboring Gwangju. Naju, the site ultimately chosen, nonetheless falls in the upper to middle range of the provincial rankings. Given the one-designation-per-province constraint and the exclusion of the most prosperous regions, namely the Seoul Metropolitan Area, Daejeon, and Chungnam, many Innovation Cities still only rank around the middle of the national distribution.

Table 1: Within Province Innovation City County Rankings

County	Wage (10k KRW)	House Prices (10k KRW/m²)	Total Employment	Manufacturing Employment	Manufacturing Share (%)
Busan	192.0	187.8	1,180,689	195,276	16.5
Daegu	176.6	160.6	770,498	155,490	20.2
Ulsan	265.0	176.6	429,500	148,160	34.5
Wonju [17]	193.9 (5)	137.0 (2)	97,791 (1)	13,687 (1)	14.0 (2)
Jincheon [11]	190.3 (4)	100.9 (3)	33,161 (5)	20,714 (3)	62.5 (1)
Eumseong [11]	187.3 (5)	87.7 (7)	43,562 (3)	24,913 (2)	57.2 (2)
Jeonju [14]	199.3 (3)	142.6 (1)	185,964 (1)	12,187 (4)	6.6 (13)
Wanju [14]	214.5 (1)	97.7 (4)	30,825 (5)	14,535 (3)	47.2 (1)
Naju [22]	181.0 (11)	84.1 (7)	26,755 (6)	5005 (5)	18.7 (7)
Gimcheon [22]	198.3 (5)	104.6 (9)	38,845 (7)	10,412 (7)	26.8 (8)
Jinju [18]	185.3 (11)	177.0 (3)	108,574 (3)	13,126 (7)	12.1 (15)
Jeju	180.5	118.3	193,721	9308	4.8
Median District	*184.1*	*102.6*	*26,038*	*5,034*	*17.5%*

Notes: Table provides within province ranking of Innovation City counties for a variety of economic indicators. Province rank in parentheses; number of districts in province in square brackets. Both omitted where the province contains only one district. Data on wages from Regional Wage Survey, on house prices from Ministry of Land, Infrastructure and Transportation, and on employment from National Business Survey. All data for the year 2010.

Once all Innovation Cities were designated, construction began on office space for the relocated agencies and residential units. Table 2 reports the final distribution of relocated public-sector workers and housing construction for each Innovation City. Relocation numbers range from 752 for the island of Jeju to 7,043 for the joint Innovation City of Jeonnam and Gwangju in Naju, totaling 42,062. Housing construction spans a similar range, from close to 2,000 units in Jeju to 19,800 in Naju. Housing units outnumber relocated jobs at every Innovation City except Bu-

san, reflecting the policy's aim of attracting residents in addition to those directly relocated.

Table 2: Extent of Innovation City Relocation

District	Project Cost in bn KRW	Relocated Workers	Housing Units
Busan	412.7	3457	2500
Daegu	1450.1	3262	7660
Ulsan	1039.0	3143	7280
Wonju (Gangwon)	839.6	6179	11,932
Jincheon & Eumseong (Chungbuk)	996.9	3116	15,184
Wanju & Jeonju (Jeonbuk)	1522.9	5300	10,299
Naju (Jeonnam)	1417.5	7043	19,800
Gimcheon (Gyeongbuk)	867.6	5561	10,686
Jinju (Gyeongnam)	1577.0	4249	13,902
Seogwipo (Jeju)	293.9	752	1840

Notes: The table provides information on the extent of the Innovation City Project for each chosen destination, regarding project cost, relocated workers, and construction effort. Cost in billion South Korean won. Project cost refers to land acquisition and office construction costs. Relocated workers gives the number of public-sector employees moved to each Innovation City. Housing units refers to newly constructed residential units accompanying the relocation. Data from MOLIT (2026).

Construction was typically concentrated at a single site, either at the outskirts or within the urban center of the designated county.[7] Aside from housing and offices, each Innovation City improved local amenities through a dedicated "Innovation Center" providing space for new businesses alongside communal facilities such as childcare centers and sports facilities.[8] The Innovation City Project thus pursued

7. The exception was Busan, which distributed its relocations across multiple districts to align institutions with existing industry structures.

8. Appendix Figure E.1 provides a visual representation of construction undertaken for the Innovation City Project.

a comprehensive development agenda at each site, aiming to create livable communities for incoming residents. Ensuring that agency staff completed the relocation required considerable effort: public-sector unions initially opposed the policy, as their members were reluctant to leave the Seoul Metropolitan Area. To this end, the policy offered relocating workers several incentives, including priority access to newly constructed housing, favorable mortgage terms or reduced lease deposits, coverage of moving expenses, job search assistance for accompanying spouses or partners, and special education programs for their children (PAPPC 2008).

3 Data

Relocation data come from the Ministry of Land, Infrastructure and Transportation (MOLIT 2026). The Ministry's website reports, for each Innovation City, the name of each relocated entity, the relocation completion date, the final number of relocated jobs, the Innovation City location, land acquisition and office construction expenditure, and the number of residential units constructed. These records are supplemented by district-level location data for each relocated entity, enabling accurate attribution of relocations to the districts within the Innovation Cities.

The remaining data stem from various sources. Employment data come from the National Business Survey, conducted by the Korean Statistical Office. The survey provides annual district-level information on employment numbers, broad industry categories, and public or private ownership for the universe of South Korean businesses. Public ownership covers two categories: corporations established under special laws and administrative bodies. The former includes corporations such as the state-owned rail and energy companies, while the latter encompasses administrative agencies such as the statistical office. State-owned corporations operate in markets, providing location-specific goods and services such as power generation and rail station maintenance. For each such corporation, headquarters are defined as public

establishments and local branches as private establishments for this study, given that local branches primarily provide local services while headquarters are more likely to supply location-independent public goods.[9] Population by age bracket and gender is available at the district level from the population census, published by the Korean Statistical Office.

Worker and resident counts by location, together with bilateral commuting patterns used in Section 7, come from the Korea Transport Institute's origin-destination commuting matrix. The Korea Transport Institute derives these volumes from commuting surveys, traffic data, and economic indicators, covering bilateral flows between and within counties for different activity types, including commuting, shopping, and leisure trips. This paper uses the commuting trip data, restricted to daily commutes defined as trips of three hours or less. The Korea Transport Institute also provides industry-level origin-destination road-freight volumes by weight and aggregate rail transport volumes, both at the county level, which are used to estimate trade elasticity in Section 6.

The model quantification also requires wages and house prices. Wages come from the Regional Wage Survey, conducted by the Korean Statistical Office, which records workers' age, hours worked, average monthly wage, industry, and residential and workplace locations. Data are available at the county level, though cities with multiple boroughs and municipalities are each aggregated to a single unit. The wage measure is the average monthly wage of working-age employees who work at least 30 hours per week. Public-sector employees are identified as those employed in the public administration sector.

9. One limitation of the establishment data is that values are censored for district-industry-year cells with fewer than three observations, with censored values reported in digit-count bins: single-digit values as *, two-digit values as **, and so on. Censored values are replaced by the average of uncensored values in the same industry–digit-count bins pairing. Censoring also obscures the ownership type of affected establishments; given the disproportionate share of private establishments, all censored observations are classified as private, except those falling in the public administration industry category. The public-sector share may therefore be marginally understated. The average annual censoring rate of 1.5% is concentrated in rural regions, leaving the sending and receiving locations largely unaffected.

Housing costs come from the Ministry of Land, Infrastructure and Transportation's real estate transaction price disclosure system, which records all real estate transactions in South Korea by property type, address, floor space, and total price. For this study county-level house prices are calculated as median price per square meter across all residential property types.

The final dataset covers bilateral travel times between counties, combining GIS data on the South Korean road and rail network from the Korea Transport Institute with train schedules from the rail operator KORAIL.[10] The paper computes separate travel time measures for freight transport and commuting. The freight measure excludes high-speed rail and caps highway speeds at the maximum permitted for trucks, while the commuting measure allows full highway speeds and travel via high-speed rail. Further detail on the travel time calculations appears in Appendix C.

4 Empirical Evidence

This section presents reduced-form evidence on the outcomes of the Innovation City Project at the Innovation Cities and their surroundings. Following Faggio (2019) and Lee, Ko, and Kim (2024), the paper uses a long-difference approach to identify the local employment multiplier of public-sector relocations for district-level employment. The estimating equation is:

$$\Delta_{2010-2018} y_{ir} = \gamma^0 \Delta_{2010-2018} N^0_{ir} + \sum_c \gamma^c \Delta_{2010-2018} N^c_{ir} + \beta X_{ir} + \Psi_r + \Delta\varepsilon_{ir}.$$

The outcome of interest, $\Delta_{2010\text{–}2018} y_{ir}$, is the change in public, total private, service, or manufacturing employment in district i in county r between 2010 and 2018. The treatment variable $\Delta_{2010\text{–}2018} N^0_{ir}$ measures the number of public-sector workers relocated to district i, while $\Delta_{2010\text{–}2018} N^c_{ir}$ measures the total number of workers relo-

10. Data were accessed via an open information request through the open disclosure system run by the Ministry of Interior and Safety.

cated to neighboring districts in distance bin c. Four distance bins of 5 km each cover all districts within 20 km of an Innovation City, with distances measured between district centroids. The control vector X_{ir} includes the change in total private employment between 2000 and 2005 to account for differential pre-designation growth trajectories. It also includes manufacturing employment share, female employment share, logged population density, squared logged population density, working-age population share, elderly population share, and total population as of 2005. Finally, the specification includes county fixed effects Ψ_r and the error term ε_{ir}.[11]

A small number of relocations target locations within the province other than the designated Innovation City. The analysis excludes these single-entity relocations to maintain consistency across treatments. The paper further restricts the sample to districts within 30 km of an Innovation City to preserve comparability between treatment and control regions, and excludes the island of Jeju owing to data limitations.

Table 3 reports the estimated number of additional employees per relocated public-sector worker in each sector. Each relocated public-sector worker raises public-sector employment by nearly one, confirming that the public employment measure accurately captures the sector despite potential value censoring. With controls, each relocated public-sector worker is associated with 1.4 additional private-sector employees, overwhelmingly in services, with a null effect on manufacturing. Estimated spillovers on total private and service employment are positive and decline with distance from the Innovation Cities, while those on manufacturing employment are marginally negative and increase with distance. None of these spillovers is statistically significant, indicating that the effects of the relocations are concentrated in the Innovation Cities themselves, with limited effects on the rest of the designated county or on surrounding counties.

11. The counties used for fixed effects are harmonized as described in Appendix B, to align with the model in the following section.

Table 3: Impact of 2010–2018 Cumulative Relocations on 2010–2018 Changes in Employment

	Public		Private					
			Total	Services	Manufacturing	Total	Services	Manufacturing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0 km	0.971***	0.952***	1.628***	1.439***	−0.032	1.439***	1.265***	−0.041
	(0.080)	(0.076)	(0.348)	(0.288)	(0.041)	(0.262)	(0.216)	(0.042)
0–5 km	−0.007	0.011	−0.204	−0.039	−0.152**	0.112	0.136	−0.044
	(0.022)	(0.019)	(0.196)	(0.160)	(0.057)	(0.152)	(0.128)	(0.038)
5–10 km	−0.014	−0.012	−0.116	−0.033	−0.053	0.008	0.040	−0.011
	(0.015)	(0.012)	(0.145)	(0.089)	(0.051)	(0.050)	(0.037)	(0.027)
10–15 km	−0.009	−0.011*	−0.010	0.005	−0.010	0.025	0.028	−0.002
	(0.012)	(0.006)	(0.103)	(0.069)	(0.042)	(0.047)	(0.032)	(0.027)
15–20 km	−0.011	−0.008	−0.067	−0.011	−0.036	−0.015	0.022	−0.024
	(0.008)	(0.006)	(0.061)	(0.037)	(0.033)	(0.047)	(0.042)	(0.023)
County FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	No	No	No	Yes	Yes	Yes
Observations	981	962	981	981	981	962	962	962
Adjusted R^2	0.636	0.680	0.106	0.116	0.033	0.412	0.436	0.173

Notes: The table reports long difference estimates on local employment numbers, dependent on relocated public-sector workers. Row (1) provides estimates on the number of additional employees in a given sector per number of relocated public-sector workers within the given sector. Row (2)–(4) provide estimates on the number of additional employees per relocated worker to districts in the respective distance bin. Columns (1)–(2) show effects on public-sector employment without and with controls, respectively. Columns (3)–(5) present estimates without controls for total, service sector, and manufacturing employment; columns (6)–(8) repeat the same specifications with controls. Controls include total employment growth 2000–2005, log population density, squared log population density, working-age share, elderly share, female employment share, manufacturing employment share, and total population. All columns include county fixed effects. Clustered standard errors at the county level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

These results exceed those of Lee, Ko, and Kim (2024), who find a public-sector multiplier on total private employment of only one for the Innovation City Project. Apart from minor differences in control variable selection and district harmonization, the discrepancy most likely reflects differences in the relocations examined and in data availability. Lee, Ko, and Kim (2024) use uncensored data, alleviating concerns about biased replacements of censored values. The authors also draw on an alternative, restricted dataset that includes many single-entity relocations outside the Innovation Cities and even outside the designated provinces. Including these additional relocations alters the control group, as more locations fall within 30 km of a relocated institution. Lee, Ko, and Kim (2024) further show that relocations to Innovation Cities have stronger effects than single-entity relocations. The present estimates should therefore be treated as an upper bound, with those of Lee, Ko, and

Kim (2024) as a lower bound.

5 Model Framework

The sectoral heterogeneity observed in the reduced-form evidence motivates a structural spatial model that examines the effects of public-sector relocation both at the Innovation Cities and on the economy as a whole in general equilibrium. Building on Moretti (2010) and Faggio (2019), this paper develops a three-sector model comprising a tradable private sector, a non-tradable private sector, and a public sector. Non-tradable goods are produced and consumed locally, while tradable goods are shipped across locations subject to trade costs. Employment and wages in the public-sector are set by the central government, which assigns workers to their workplace locations. Labor is the only factor of production, and developable land, fixed in local supply, is used for housing. Households derive utility from housing, tradable goods, non-tradable goods, and the public good, and hold idiosyncratic preferences over their residence and workplace locations.

Following Moretti (2010) and Faggio (2019), the model allows for free migration between locations. Unlike their closed-city frameworks, however, this paper also incorporates commuting, allowing households to reside in a different location from their workplace. Commuting spreads demand effects across nearby locations, with increased commuting times reducing household utility. Kim and Hong (2019) document that roughly a fifth of South Korea's population commutes outside its residential county for work. They further show that service-sector multiplier effects from new manufacturing jobs accrue disproportionately at the residence rather than the workplace location, underscoring the relevance of commuting in the South Korean context.

Unlike Faggio (2019), this paper explicitly assumes that relocated jobs are filled by relocated workers rather than local hires. The model treats public and private

sector workers as distinct types, each confined to their own sector. Workers therefore cannot switch sectors in response to potentially higher wages. The public-sector workers are furthermore directly assigned to their workplace by the government. Jofre-Monseny, Silva, and Vázquez-Grenno (2020) develop an alternative approach with homogeneous workers that examines the effect of the public sector on local wages.

To isolate the effect of public-sector workers on the local economy, the model treats public-sector output as a homogeneous, freely traded, non-competitive public good produced with equal productivity across all locations. Since the total number of public-sector workers is fixed, aggregate public-sector output remains constant regardless of how workers are distributed across locations.

A public-sector relocation increases the number of workers at the target location. Since workers can commute, relocated workers may choose to reside in the target location or in a nearby district. These new residents raise local demand for housing, tradable goods, and non-tradable goods, pushing up prices for all three. Rising prices induce some existing residents to leave, accepting higher commuting costs or lower amenities to avoid the higher cost of living. Higher non-tradable prices translate directly into higher wages in that sector, as labor is its only input and demand is fulfilled locally. In the tradable sector, however, higher local prices make the location a more attractive market for goods produced elsewhere. Since workers are mobile between private sectors, the tradable sector must match non-tradable wages, raising its production costs and undermining its competitiveness.

Two factors counteract this price pressure. A net increase in workers at the target location raises local productivity through agglomeration economies (Combes et al. 2012), increasing output per worker and lowering unit production costs. A larger resident population in the target location and surrounding regions also confers a home market advantage on local producers (Krugman 1980). With more consumers nearby, a greater share faces low trade costs for locally produced goods. The net

effect on the tradable sector is therefore indeterminate.

5.1 Workers

The model extends Monte, Redding, and Rossi-Hansberg (2018) by incorporating a public sector and a non-tradable service sector. The economy comprises a set of locations $n, i \in \mathcal{N}$ and two sets of households: those in the public-sector workforce $\bar{L}^G$ and those in the private-sector workforce $\bar{L}^P$. Both worker types share the same utility function $U_{ni\omega}$, deriving utility from tradable consumption C_{nT}, non-tradable consumption C_{nN}, housing H_n, and a public good V. Utility is augmented by idiosyncratic workplace-residence preferences $b_{ni\omega}$ and diminished by a commuting disutility κ_{ni} that increases with travel time.

$$U_{ni\omega} = \frac{b_{ni\omega}}{\kappa_{ni}} \left(\frac{C_{nT}}{\alpha_T}\right)^{\alpha_T} \left(\frac{C_{nN}}{\alpha_N}\right)^{\alpha_N} \left(\frac{H_n}{1-\alpha_T-\alpha_N}\right)^{1-\alpha_T-\alpha_N} V. \tag{1}$$

Each worker ω draws idiosyncratic amenities $b_{ni\omega}$ from a Fréchet distribution with scale parameter $B_{ni} > 0$ and shape parameter $\varepsilon > 1$. The scale parameter B_{ni} captures the baseline attractiveness of any workplace-residence pair, independent of commuting time; commuting time enters utility separately through the disutility κ_{ni}. The shape parameter ε governs the heterogeneity of idiosyncratic preferences: a higher value implies more homogeneous worker tastes, raising the relevance of local prices and wages for location choices.

$$G_{ni}(b) = \Pr[b_{ni\omega} \leq b] = \exp\{-B_{ni} b^{-\varepsilon}\}. \tag{2}$$

The worker index ω is suppressed henceforth, since all workers in the same sector and residence-workplace combination behave identically, earning the same wage and making the same consumption decisions. Following Monte, Redding, and Rossi-Hansberg (2018), tradable consumption takes the form of a constant elasticity of substitution (CES) composite over good varieties q. The elasticity of substitution

σ governs how readily consumers substitute between varieties: higher values reduce firms' pricing power and increase consumer sensitivity to price differences. Each worker chooses the optimal quantity of each variety across all regions i:

$$C_{nT} = \left[\sum_{i \in N} \int_0^{M_i} c_{niT}(q)^{\rho} \, dq \right]^{1/\rho}, \qquad \sigma = \frac{1}{1-\rho} > 1. \tag{3}$$

5.2 Housing Sector

All developable land $\bar{H}$ is owned by local landlords, who rent out floor-space H_n at price Q_n and spend their income on tradable and non-tradable goods. Their expenditure shares on tradable and non-tradable goods are $\frac{\alpha_T}{\alpha_N+\alpha_T}$ and $\frac{\alpha_N}{\alpha_N+\alpha_T}$, respectively. Land development increases in local housing prices with elasticity δ, yielding the floor-supply schedule $H_n = \bar{H}_n Q_n^{\delta}$. Total housing expenditure equals the combined spending of public- and private-sector households:

$$Q_n H_n = (1 - \alpha_N - \alpha_T) \sum_{J \in \{P,G\}} v_n^J R_n^J, \tag{4}$$

where v_n^J gives average residential income of J-type, either public G or private P, households in location n, with R_n^J giving the number of J-type households in location n. With housing market clearing, the equilibrium price per unit of floor-space can be written as:

$$Q_n = \left[(1 - \alpha_T - \alpha_N) \frac{\sum_{J \in \{P,G\}} v_n^J R_n^J}{\bar{H}_n} \right]^{\frac{1}{1+\delta}}. \tag{5}$$

5.3 Goods

Total expenditure on tradable and non-tradable goods combines expenditure of the private and public households, as well as the expenditure of the landlords on the respective good:

$$
\begin{aligned}
P_{nN}C_{nN} &= \alpha_N \sum_{J\in\{P,G\}} v_n^J R_n^J + \frac{\alpha_N}{\alpha_N+\alpha_T}(1-\alpha_T-\alpha_N) \sum_{J\in\{P,G\}} v_n^J R_n^J \\
&= \frac{\alpha_N}{\alpha_N+\alpha_T} \sum_{J\in\{P,G\}} v_n^J R_n^J.
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
P_{nT}C_{nT} &= \alpha_T \sum_{J\in\{P,G\}} v_n^J R_n^J + \frac{\alpha_T}{\alpha_N+\alpha_T}(1-\alpha_T-\alpha_N) \sum_{J\in\{P,G\}} v_n^J R_n^J \\
&= \frac{\alpha_T}{\alpha_N+\alpha_T} \sum_{J\in\{P,G\}} v_n^J R_n^J.
\end{aligned} \tag{7}
$$

In Eq. (6) and Eq. (7), P_{nN} and P_{nT} describe the respective price of the tradable and the non-tradable good at location n. The non-tradable sector produces output under perfect competition according to $C_{nN} = A_n L_{nN}^P$, where A_n denotes labor productivity and L_{nN}^P denotes labor input in location n. Productivity comprises an exogenous component $\bar{A}_i$ that scales with total local employment with elasticity ν, yielding aggregate productivity

$$
A_n = \bar{A}_n (L_{nN}^P + L_{nT}^P + L_n^G)^\nu. \tag{8}
$$

Public- and private-sector workers are perfect substitutes, with no sector-specific productivity effects.[12] Under perfect competition, the non-tradable unit price is:

$$
P_{nN} = \frac{w_n^P}{A_n}. \tag{9}
$$

The private-sector wage w_n^P applies to all private-sector employees. Private-sector households freely switch between sectors within a region, equalizing wages between

12. Evidence on the productivity effects of administrative institutions is sparse, with existing studies finding negligible or slightly negative effects on productivity-related indicators such as value added per worker (Becker, Heblich, and Sturm 2021; Seo and Do 2023). The existing evidence for public-private complementarities relates primarily to research-oriented institutions such as universities or dedicated research institutes (Jaffe 1989; Abramovsky, Harrison, and Simpson 2007; Andersson, Quigley, and Wilhelmsson 2009). Since the majority of relocated institutions are administrative rather than research-focused, this paper models productivity through conventional agglomeration economies following Combes et al. (2012). For an alternative specification treating public and private workers as complements and incorporating regional productivity spillovers, see Freitas (2025).

them. Combining the non-tradable supply condition with Eq. (6) yields the market-clearing condition in the non-tradable sector:

$$w_n^P L_{nN}^P = \frac{\alpha_N}{\alpha_N + \alpha_T} \sum_{J \in \{P,G\}} v_n^J R_n^J. \tag{10}$$

The tradable sector operates under monopolistic competition. As in the non-tradable sector, labor is the only input and output depends on local productivity. A given variety q produced in location i and consumed in n is priced at:

$$p_{niT}(q) = \frac{\sigma}{\sigma - 1} \frac{\tau_{ni} w_i^P}{A_i}. \tag{11}$$

The price reflects the unit cost $\frac{w_i^P}{A_i}$, scaled by iceberg trade costs τ_{ni}, which raise the price at the location of consumption. Each variety q is produced by a single firm that pays fixed labor costs F upon market entry, yielding the production function $x_i(q) = A_i l_i - F$, where l_i denotes firm-level labor input. Combining the profit-maximization condition with the zero-profit condition yields $x_i(q) = A_i F(\sigma - 1)$, which, combined with the labor market-clearing condition, gives the number of firms in location i as:

$$M_i = \frac{L_{Ti}^P}{\sigma F}. \tag{12}$$

Total private-sector employment in the tradable sector of location i is denoted L_{iT}^P. Trade shares between locations follow a gravity structure:

$$\pi_{ni} = \frac{M_i {p_{niT}}^{1-\sigma}}{\sum_k M_k {p_{nkT}}^{1-\sigma}} = \frac{L_{iT}^P \left(\tau_{ni} w_i^P / A_i\right)^{1-\sigma}}{\sum_k L_{kT}^P \left(\tau_{nk} w_k^P / A_k\right)^{1-\sigma}}. \tag{13}$$

The share of tradable goods produced in i and consumed in n rises with the number of local firms M_i and falls with the local sales price p_{niT}, relative to firm numbers and prices elsewhere. Given the zero-profit condition and balanced trade across all locations, tradable-sector income in location i equals total expenditure across all regions on goods produced in i:

$$w_i^P L_{iT}^P = \sum_{n \in N} \pi_{ni} \frac{\alpha_T}{\alpha_N + \alpha_T} \sum_{J \in \{P,G\}} v_n^J R_n^J. \tag{14}$$

The tradable price index follows from combining individual good prices from Eq. (11), the number of firms from Eq. (12), and trade shares from Eq. (13):

$$\begin{aligned} P_{nT} &= \left[\sum_{i \in N} \int_0^{M_i} p_{nT}(q)^{1-\sigma} \, dj \right]^{\frac{1}{1-\sigma}} \\ &= \frac{\sigma}{\sigma - 1} \left(\frac{1}{\sigma F} \right)^{\frac{1}{1-\sigma}} \left[\sum_i L_{Ti}^P \left(\frac{\tau_{ni} w_i^P}{A_i} \right)^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \\ &= \frac{\sigma}{\sigma - 1} \left(\frac{L_{nT}^P}{\sigma F \, \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{\tau_{nn} w_n^P}{A_n}. \end{aligned} \tag{15}$$

In addition to the three sectors that directly enter household utility, the economy includes a government sector. The public sector produces a non-competitive, freely tradable public good V using only labor, with equal productivity across all locations. Public-good output and its distribution are invariant to the spatial allocation of workers, provided the total number of public-sector workers remains fixed at $\sum_i L_i^G = \bar{L}^G$; the public good is therefore suppressed henceforth. Public-sector workers are assigned their workplace by the central government and receive a uniform wage w^G, irrespective of workplace location. The wage bill is financed by a flat income tax t on all workers:

$$\sum_i w_i^G L_i^G = t \times \sum_{J \in \{G,P\}} \sum_i w_i^J L_i^J. \tag{16}$$

5.4 Location Choice

The indirect utility function of a household of type J follows from substituting the consumption prices in Eq. (9), Eq. (5), and Eq. (15):

$$U_{ni}^J = \frac{B_{ni} \, w_i^J (1-t)}{\kappa_{ni} \, {P_{nT}}^{\alpha_T} \, {P_{nN}}^{\alpha_N} \, {Q_n}^{1-\alpha_T-\alpha_N}}. \tag{17}$$

Households in both sectors prefer workplaces with high net wages $(1-t)w_i^J$, though

only private-sector households face wage variation across counties. Households also prefer residence locations with low tradable and non-tradable prices and low housing costs. Additionally, they maximize utility by selecting residence-workplace pairs with short travel times and high amenity values. Public-sector workers have their workplace assigned by the central government and therefore optimize only their residential location, conditional on that assignment. Private-sector workers, by contrast, freely choose both workplace and residence to maximize utility. Since workers draw amenities from a Fréchet distribution, the share of private-sector workers choosing residence n and workplace i follows from the probability that the pair (n, i) yields the highest utility among all alternatives. The resulting expression takes the following gravity structure:

$$\lambda_{ni}^{P} = \frac{B_{ni}\Big(\kappa_{ni} {P_{nT}}^{\alpha_T} {P_{nN}}^{\alpha_N} Q_n^{1-\alpha_T-\alpha_N}\Big)^{-\varepsilon}\Big[w_i^P(1-t)\Big]^{\varepsilon}}{\sum_r \sum_s B_{rs}\Big(\kappa_{rs} {P_{rT}}^{\alpha_T} {P_{rN}}^{\alpha_N} Q_r^{1-\alpha_T-\alpha_N}\Big)^{-\varepsilon}\Big[w_s^P(1-t)\Big]^{\varepsilon}}. \tag{18}$$

The population shares, conditioned on workplace or residence location, distribute private workers $L_i^P = L_{iT}^P + L_{iN}^P$ and residents across locations. Summing over all residence locations recovers total private employment at each workplace, while summing over all workplace locations recovers total residents at each location:

$$L_i^P = \sum_n \lambda_{ni|n}^P \, R_n^P, \tag{19}$$

$$R_n^P = \sum_i \lambda_{ni|i}^P \, L_i^P. \tag{20}$$

Residential income for private-sector workers in location n follows from the conditional shares $\lambda_{ni|n}^P$ summed over all workplace locations, net of income tax:

$$\bar{v}_n^P = (1-t) \sum_i \lambda_{ni|n}^P \, w_i^P. \tag{21}$$

Since public- and private-sector households share identical utility functions, the

conditional residential shares depend only on residential location characteristics and apply equally to public-sector households:

$$\lambda^G_{ni|i} = \lambda^P_{ni|i} = \lambda_{ni|i} = \frac{B_{ni}\Big(\kappa_{ni} {P_{nT}}^{\alpha_T} {P_{nN}}^{\alpha_N} {Q_n}^{1-\alpha_T-\alpha_N}\Big)^{-\varepsilon}}{\sum_r B_{ri}\Big(\kappa_{ri} {P_{rT}}^{\alpha_T} {P_{rN}}^{\alpha_N} {Q_r}^{1-\alpha_T-\alpha_N}\Big)^{-\varepsilon}}. \tag{22}$$

Conditional private-sector residential choices therefore yield public-sector resident counts, analogously to Eq. (20):[13]

$$R^G_n = \sum_i \lambda_{ni|i}\, L^G_i. \tag{23}$$

With free mobility, expected utility equalizes across all private-sector residence-workplace pairs:

$$\bar{U}^P = E[U^P_{ni\omega}] = \Gamma\left(\frac{\varepsilon-1}{\varepsilon}\right)\left[\sum_r \sum_s B_{rs}\Big(\kappa_{rs} {P_{rT}}^{\alpha_T} {P_{rN}}^{\alpha_N} {Q_r}^{1-\alpha_T-\alpha_N}\Big)^{-\varepsilon}\Big(w^P_s(1-t)\Big)^{\varepsilon}\right]^{1/\varepsilon}. \tag{24}$$

When utility differs across locations, households migrate toward higher-utility pairs until wage and price adjustments restore equality, as migration is costless. Expected utility combines the location-pair components of the indirect utility function, scaled by the Gamma function $\Gamma\left(\frac{\varepsilon-1}{\varepsilon}\right)$.

5.5 Equilibrium

The equilibrium is characterized by nine variables:

$$\Big\{ w^P_n,\ v^P_N,\ L^{PN}_n, L^{PT}_n,\ R^P_n,\ R^G_n,\ P_{nT},\ P_{nN},\ Q_n \Big\}_{n=1}^{N},$$

two scalars t and $\bar{U}^P$, and ten equations. All goods markets clear: the tradable goods market in Eq. (14), the non-tradable market in Eq. (10), and the housing market in

13. Recovering $\bar{v}^G_n$ is not required in the present model, but follows from dividing total residential wage income by the total number of residents: $\bar{v}^G_n = \frac{(1-t)\sum_i \lambda^G_{ni|i} w^G_i L^G_i}{\sum_i \lambda^G_{ni|i} L^G_i}$.

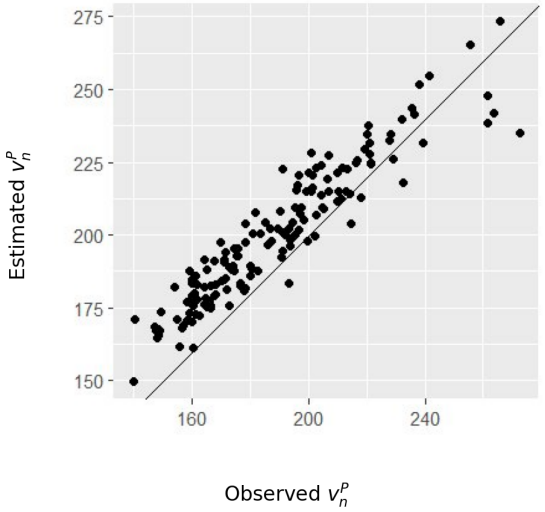
Eq. (5). Additionally, the model requires public- and private-sector residence choices in Eq. (23) and Eq. (20), private-sector workplace choices in Eq. (19), and average private-sector residential income from Eq. (21). Eq. (15) and Eq. (9) determine the tradable and non-tradable price indices, respectively, while government budget equalization in Eq. (16) determines the tax rate. All remaining endogenous variables follow from combining these variables and equations.

6 Calibration

Calibration begins by recovering residents and residential income by type for each location. Public-sector employment shares are observed from the National Business Survey data and then applied to the county-level worker numbers in the commuting matrix of the Korea Transport Institute. Residential counts follow from conditioning commuting shares on workplace location and multiplying by employment at each location and in each sector: $R_n^J = \sum_i \lambda_{ni|i}^J L_i^J$. Residential income requires the tax rate t, recovered from the balanced government budget in Eq. (16), yielding an income tax rate of 12.3 %. Since government wages are uniform across space, public-sector residential income is $\bar{v}^G = (1-t)w^G$ in all locations. Private-sector residential income follows as $\bar{v}_n^P = \sum_i \lambda_{ni|n}^P w_i^P (1-t)$. Figure 2 compares estimated and observed private sector income; a limited average deviation of 7.4 % confirms that the model fits the data reasonably well.

Housing fundamentals $\bar{H}_n$ follow from Eq. (5), combining house price data with the residential income recovered above. Calibration also requires household expenditure shares. Using Consumer Price Index item weights from the Bank of Korea, published via the OECD for 2010 (OECD 2026), South Korean households allocate 39 % of consumption expenditure to goods, treated as a proxy for tradables, and 44 % to services, treated as a proxy for non-tradables, leaving 17 % for housing. The expenditure shares are therefore $\alpha_T = 0.39$ and $\alpha_N = 0.44$. The housing supply

Figure 2: Residential Wage Moment



Notes: Fit between model estimated and observed residential income by county. Both estimated and observed income show pre-tax values. Observed residential income from Regional Wage Survey.

elasticity is set to $\delta = 1.71$, following J. Lee (2024).

Since all components of the balanced trade condition in Eq. (14) are observed except productivity A_i, the model solves the system of N equations jointly. The solution yields productivities $\{A_i\}$ that ensure balanced trade given observed values of $\bar{v}_n^J$, R_n^J, w_i, L_i^T, and d_{ni}^f. Productivity estimation requires setting the trade elasticity σ and parameterizing trade costs. The trade elasticity is set to $\sigma = 4$, following Broda and Weinstein (2006). Trade costs are parameterized following Head and Mayer (2014) as a function of freight network travel times: $\tau_{ni} = (d_{ni}^f)^{\psi} e^{\beta \mathbf{C}} \tilde{e}_{ni}$, where ψ is the trade cost elasticity with respect to travel time along the freight network d_{ni}^f, $\mathbf{C}$ is a vector of controls, and $\tilde{e}_{ni}$ is an error term.

To estimate the trade elasticity ψ, the gravity trade equation $X_{ni} = \sum_{n \in N} \pi_{ni} \frac{\alpha_T}{\alpha_N + \alpha_T} \frac{\bar{v}_n \bar{R}_n}{P_{nT}}$ is log-linearized. As data are available only on the volume of bilateral trade, not its value, the equation is first transformed into bilateral trade volumes by dividing by the price p_{niT} (Eq. (11)):

$$\begin{aligned}\ln(K_{ni}) &= \ln(L_{iT}) - \sigma \ln(w_i^P / A_i) - \sigma \ln(\tau_{ni}) \\ &\quad + (\sigma - 1) \ln(\Omega_n) + \ln\left(\frac{\alpha_T}{\alpha_N + \alpha_T} \frac{\bar{v}_n \bar{R}_n}{P_{nT}}\right) \\ &= \zeta_n + \chi_i - \sigma \ln(\tau_{ni}),\end{aligned} \tag{25}$$

where K_{ni} denotes the volume of goods shipped from i to n and $\Omega_n = \sum_k L_{kT}^P \left(\tau_{nk} w_k^P / A_k\right)^{1-\sigma}$ gives the aggregate imports from all locations $k \in N$ into location n. All components except the trade cost τ_{ni} are absorbed into the origin and destination fixed effects ζ_n and χ_i. Substituting the parameterization of τ_{ni} yields the estimating equation:

$$ln(K_{ni}) = \zeta_n + \chi_i - \sigma\psi ln(d_{ni}^f) - \sigma\beta\mathbf{C} - \sigma ln(\tilde{e}_{ni}). \tag{26}$$

The analysis uses commodity flow data from the Korea Transport Institute to estimate Eq. (26). Since these data are available only for select years, the analysis uses 2011 values for both freight volume and travel times, the year closest to the remaining data sources. Road freight is disaggregated by sector, while rail freight is available as total volume only, with flows recorded in tonnes. The analysis combines road and rail freight to obtain total freight volume and aggregates the result to the harmonized district level. The sample excludes Jeju, as only road and rail transport data are observed there, leaving shipping flows to and from the island unobservable.

The specification addresses measurement limitations and infrastructure endogeneity through controls and an instrumental variable strategy. Because shipping data are unavailable, the specification includes a dummy for coastal district pairs. This corrects for potential under-reporting of coastal trade, even though shipping constitutes only 7 % of domestic South Korean freight (KTI 2011). A control for cross-provincial pairs addresses potential discontinuities at provincial borders. A further concern is the non-random placement of infrastructure: roads and railways may have been built to serve already active trade routes. Travel time is therefore

Table 4: Gravity Trade Estimation

	(1) OLS	(2) IV	(3) OLS	(4) IV
Dependent variable: ln(Trade Volume)				
ln(Travel Time)	-0.990^{***}	-0.998^{***}	-1.205^{***}	-1.247^{***}
	(0.040)	(0.040)	(0.023)	(0.024)
Different State	−0.094	−0.086	−0.054	−0.008
	(0.065)	(0.065)	(0.057)	(0.057)
Coastal Pair	0.122^{***}	0.120^{***}	0.009	−0.001
	(0.040)	(0.041)	(0.014)	(0.015)
Sector FE	No	No	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Observations	23,716	23,716	443,876	443,876
R^2	0.894	0.894	0.631	0.631

Notes: The table shows OLS and IV estimates of trade elasticities. Row (1) provides the percentage change in bilateral freight volume resulting from a 1 % increase in travel time between counties. Rows (2) and (3) control for province border crossing and coastal status of the county pair. Columns (1) and (2) show results for aggregate shipment volume, Columns (3) and (4) show results for volume by sector. Aggregate shipment includes rail transport, with sectoral shipments only counting road freight. Columns (1) and (3) show OLS estimates, Columns (2) and (4) IV estimates. Origin and destination county two-way clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

instrumented with bilateral distances between district centroids, as an alternative to standard OLS estimation. The estimation proceeds in two specifications: one for total freight volume including rail and one for sectoral freight volume using road traffic only.

Table 4 reports the results. Depending on the specification, the coefficient ranges from −0.990 to −1.247. Given $\sigma = 4$, this implies a range for ψ from 0.25 to 0.31. This paper sets ψ to 0.31, following column (4). The trade cost elasticity is therefore lower than the value of $\psi = 0.43$ that Monte, Redding, and Rossi-Hansberg (2018) find for the United States.

With trade costs parameterized as $\tau_{ni} = (d_{ni}^f)^{\psi}$, aggregate productivity for each location follows from the balanced trade condition in Eq. (14) as described above. The agglomeration elasticity is set to $\nu = 0.05$, the value estimated for South Korea by J. K. Lee (2021), consistent with the literature surveyed by Rosenthal and Strange

(2004). Exogenous productivities then follow as $\bar{A}_i = A_i / L_i^{\nu}$.

Bilateral amenities B_{ni} are recovered next. As with productivities, they follow from the commuting gravity equation in Eq. (18), but require two parameters: the shape parameter ε and the commuting disutility κ. As with trade costs, commuting disutility is parameterized as a power function of travel times, $\kappa_{ni} = d_{ni}^{\mu}$, where μ is the commuting cost elasticity with respect to travel time and d_{ni} denotes travel times across the South Korean road and rail network, including high-speed rail. Log-linearizing the commuting share equation yields the following equation, after replacing the prices with Eq. (15), (9) and (5):

$$\lambda_{ni}^{P} = \frac{B_{ni} d_{ni}^{-\mu\varepsilon} \left(\frac{L_{nT}^{P}}{\pi_{nn}}\right)^{\frac{-\alpha_T \varepsilon}{1-\sigma}} \left(\frac{\tau_{nn} w_n^P}{A_n}\right)^{-\alpha_T \varepsilon} \left(\frac{w_n^P}{A_n}\right)^{-\alpha_N \varepsilon} \left(\frac{\bar{v}_n \bar{R}_n}{H_n}\right)^{\frac{-(1-\alpha_T-\alpha_N)\varepsilon}{1+\delta}} w_i^{P\varepsilon}}{\sum_r \sum_s B_{rs} d_{rs}^{-\mu\varepsilon} \left(\frac{L_{Tr}^{P}}{\pi_{rr}}\right)^{\frac{-\alpha_T \varepsilon}{1-\sigma}} \left(\frac{\tau_{rr} w_r^P}{A_r}\right)^{-\alpha_T \varepsilon} \left(\frac{w_r^P}{A_r}\right)^{-\alpha_N \varepsilon} \left(\frac{\bar{v}_r \bar{R}_r}{H_r}\right)^{\frac{-(1-\alpha_T-\alpha_N)\varepsilon}{1+\delta}} w_s^{P\varepsilon}}, \tag{27}$$

$$ln(\lambda_{ni}^{P}) = g_0 + \eta_n + \varepsilon ln(w_i^P) - \mu\varepsilon ln(d_{ni}) + u_{ni}, \tag{28}$$

where all price components are absorbed into the origin fixed effect η_n, the bilateral amenity B_{ni} is absorbed into the error term u_{ni}, and g_0 is a constant. The workplace wage w_i^P and bilateral travel time d_{ni} enter separately. Table 5 reports OLS estimates of $\varepsilon = 4.01$ and $\mu\varepsilon = 3.99$, yielding $\mu = 1.00$. The commuting elasticity exceeds the distance elasticity for trade, confirming that goods are more mobile than people.

With ε and μ recovered, bilateral amenities follow from Eq. (27) and the commuting share data.

7 Counterfactual Analysis

This paper uses the calibrated model to run two related counterfactuals examining the effects of the Innovation City Project's public-sector relocations on the Innova-

Table 5: Gravity Commuting Estimation

	(1) OLS
Dependent variable: ln(*Commuting Share*)	
ln(Wage)	4.011^{***}
	(0.030)
ln(Travel Time)	-3.992^{***}
	(0.129)
Origin FE	Yes
Observations	22,049
R^2	0.622

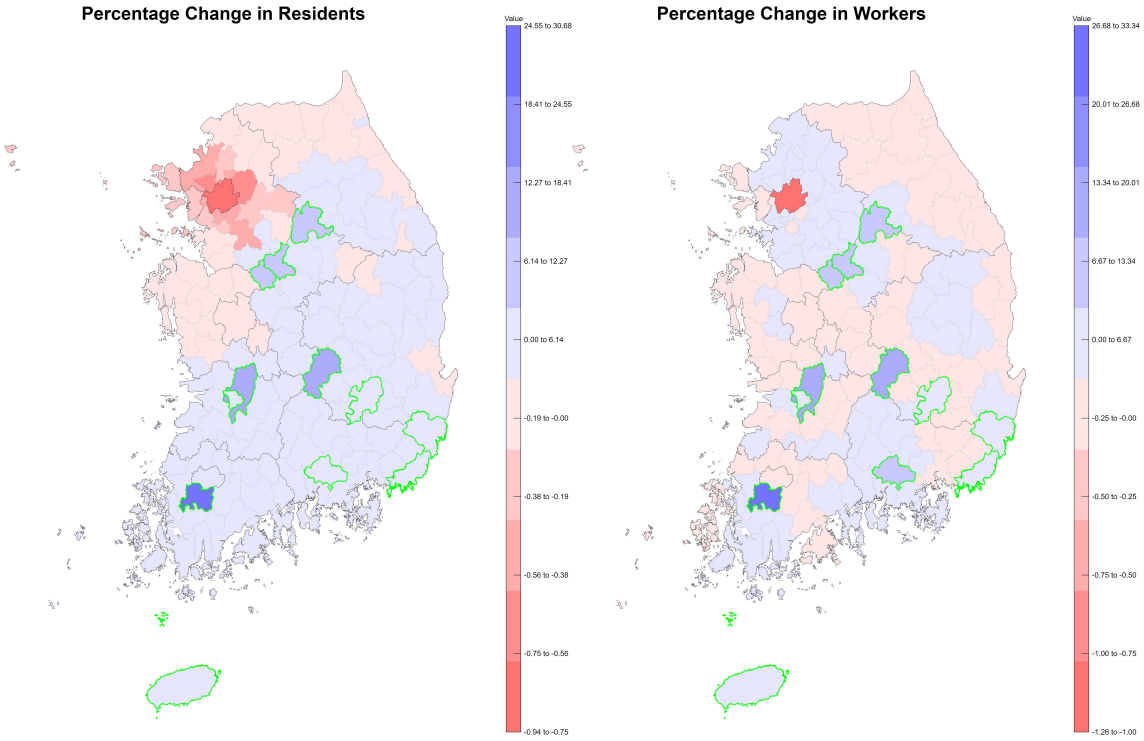
Notes: The table provides OLS estimates of commuting elasticities. Row (1) provides the percentage change in bilateral commuting shares resulting from a 1 % increase in wages at the destination. Row (2) provides the percentage change in bilateral commuting shares resulting from a 1 % increase in travel times between origin and destination. Origin county clustered standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

tion Cities, the Seoul Metropolitan Area, and the wider South Korean economy.[14] The first counterfactual simulates the relocations and the accompanying housing construction, modeling the additional housing as an increase in land supply using Eq. (5). For each Innovation City, the calibration sets the exogenous land supply $\bar{H}_n$ to maintain the observed baseline rent level Q_n after increasing the local resident population by the number of constructed residential units. For cross-border Innovation Cities, only aggregate housing totals are observed; the district share of relocated jobs is used to proxy for the share of housing allocated to each district.

The analysis focuses on the policy's stated goals: the movement of residents and jobs out of the Seoul Metropolitan Area, and out of Seoul in particular, into the Innovation Cities. Since the policy also aimed to improve productivity and attract industry to the Innovation Cities, the analysis examines effects on sectoral employment in the non-tradable and tradable sectors.

14. Algorithms for calculating the counterfactuals and estimating amenities, productivities, and other supplementary variables appear in Appendix A.

Figure 3: Resident and Worker Changes of Counterfactual Innovation City Project



Notes: Percentage Changes in county-level residents and workers post counterfactual Innovation City Project, including public-sector relocation and housing construction. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red. Left residents, right total employment.

7.1 Employment Effects

Figure 3 shows the results for population and total employment. Relocating roughly 42,000 public-sector workers produces meaningful population changes, especially for the initially smaller provincial Innovation Cities, with some in the southwest experiencing population increases of up to 31 %. These increases result mainly from population declines in the Seoul Metropolitan Area, particularly from Seoul itself. Provinces south of the Seoul Metropolitan Area without an Innovation City designation show marginal declines in residents, as do some remote counties in the northeastern region.

Net employment changes reveal a direct link between outflows from the Seoul Metropolitan Area and inflows to the Innovation Cities. Regions outside the Seoul Metropolitan Area other than the Innovation Cities experience only minor net em-

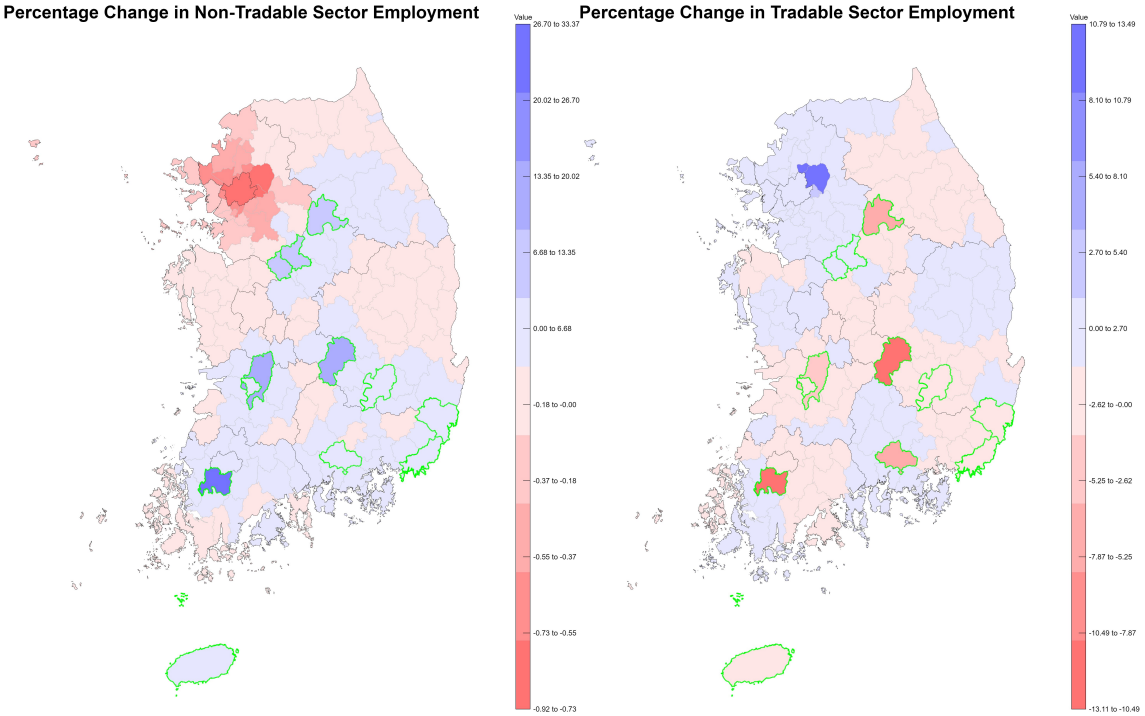
ployment changes, losing 900 workers in aggregate. By contrast, the Seoul Metropolitan Area loses 57,000 workers, entirely absorbed by the Innovation Cities. These net effects, however, mask heterogeneity across regions. Within the Seoul Metropolitan Area, Seoul, its nearby satellite cities, and Incheon account for the entire employment decline, while all other locations register small gains. Among locations outside the Seoul Metropolitan Area, counties farther from the Innovation Cities that retain their resident population register small employment gains, while the remaining counties register small employment losses.

Because households can commute to work, residential changes differ from employment changes. Most counties in the Seoul Metropolitan Area lose residents despite gaining employment, as Seoul households find employment outside the city. The pattern reverses around the Innovation Cities: nearby counties lose employment but gain residents, as some workers employed at the Innovation Cities choose to reside in neighboring counties and commute. Of the net outflow of 56,000 residents from the Seoul Metropolitan Area, roughly 10 % move to destinations other than the Innovation Cities.

At first glance, the public-sector relocation appears to fulfill the main goals of the policy, as people and jobs move from the Seoul Metropolitan Area to the Innovation Cities. Of the 57,000 jobs that leave Seoul, however, 42,000 are the public-sector relocations themselves, indicating that the relocated public-sector jobs generate little additional private employment. On average, each relocated public-sector job creates only 0.38 private-sector jobs.

The limited multiplier reflects opposing dynamics in the non-tradable and tradable private sectors, each of which the analysis examines in turn. Figure 4 shows changes in private non-tradable and tradable employment. Non-tradable employment largely follows the pattern of residential change: rising county population increases local spending on non-tradables and therefore employment in the sector. One divergence between residents and non-tradable workers appears in several central

Figure 4: Private Non-Tradable and Tradable Employment Changes of Counterfactual Innovation City Project



Notes: Percentage Changes in county-level non-tradable and tradable private employment post counterfactual Innovation City Project, including public-sector relocation and housing construction. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red. Left non-tradable employment, right tradable employment.

eastern counties, which show marginally positive residential growth but a marginal decline in non-tradable employment. This divergence stems from a marginal wage increase across locations outside the Seoul Metropolitan Area (see Appendix Figure E.4), which counteracts the rise in local spending.

Tradable-sector changes run broadly inverse to those in the non-tradable sector. The inflow of public-sector workers raises the population at the Innovation Cities, driving up demand for goods and housing and bidding up private-sector wages and rents. Despite these demand pressures, the accompanying construction effort keeps rents below their baseline level overall; private-sector wages at the Innovation Cities, however, rise by up to 1.6 % (Appendix Figure E.4). Although the Innovation Cities also record productivity gains from increased local employment, the rising labor costs outweigh these gains, leaving the local tradable sector less competitive and causing it to contract. A related mechanism operates in the surrounding regions, where private-sector wages also rise because of higher local rents, if only marginally. A concurrent decline in local productivity compounds this wage increase, further reducing the regions' attractiveness to the tradable sector. The tradable sector therefore contracts across both the Innovation Cities and the surrounding regions.

Two locations gain tradable employment despite the general contraction: the neighboring Innovation City counties of Eumseong and Jincheon. Both have high underlying productivities and high ratios of housing construction to relocated workers, generating stronger productivity effects while limiting pressure on local housing prices. Even under these favorable conditions, tradable-sector gains remain marginal: the sector increases employment by only around 1.1 % and 0.7 % for Eumseong and Jincheon respectively. The other locations with the lowest outflow of tradable-sector employment are Ulsan and Jeju, both with outflows below 0.2 %. The limited outflow in Ulsan follows the same reasoning as Eumseong and Jincheon, with high underlying productivity and housing-to-relocation ratios. Jeju has neither of these advantages. Its island status, however, confers a home market advantage on the lo-

cal tradable sector, as all non-local producers face high trade costs when supplying local residents.

The clear beneficiary is the tradable sector in the Seoul Metropolitan Area. Private-sector wages decline as population falls, due to reduced rent pressure, while rising employment strengthens local productivity. This shift reinforces itself: a larger tradable sector in the Seoul Metropolitan Area reduces trade costs for local residents, since a greater share of consumption can be sourced locally. Households therefore accept lower wages in exchange for cheaper consumption goods. Seoul itself records the largest outflow of workers. The resulting productivity decline makes the city less attractive to the tradable sector than neighboring counties, yielding a smaller employment gain.

Across the Innovation Cities, non-tradable employment gains more than offset tradable losses, yielding a positive private employment multiplier. In aggregate, the Innovation Cities lose roughly 10,500 tradable jobs to the Seoul Metropolitan Area, with surrounding regions losing a further 3,500 on net. Against these losses, the Innovation Cities gain 26,200 non-tradable-sector jobs, with a further 2,700 created in the remaining counties outside the Seoul Metropolitan Area. The estimates imply public-sector multipliers of 0.62 for non-tradable employment and -0.24 for tradable employment, combining to yield the private employment multiplier of 0.38.

The aggregate multiplier masks meaningful heterogeneity across locations, already indicated by the heterogeneous effects on tradable employment. Innovation Cities with low multipliers tend to have low housing-to-relocated-worker ratios as well as low underlying productivities relative to neighboring regions. Gimcheon builds only two residential units per relocated public-sector worker and borders the county of Gumi, which is significantly more productive, resulting in a limited multiplier of 0.27. Eumseong and Jincheon, two Innovation Cities with multipliers close to 0.9, have higher productivities than their neighbors and construct close to five residential units per relocated worker. Jeju also shows a relatively high multiplier

of 0.7, reflecting its island status. Since commuting from Jeju to the mainland is impractical, relocated workers reside on the island, generating stronger demand for local services than at other Innovation Cities. The tradable sector additionally benefits from a home market advantage, as competing producers face high trade costs when supplying Jeju's residents.

7.2 Welfare Effects

Following up on the changes in employment, welfare of private sector workers is calculated and compared to the baseline using Eq. (24). Private-sector utility increases by less than 0.04 %, indicating negligible welfare gains. These estimates, however, exclude welfare effects on public-sector workers. Since public-sector workers lack free workplace choice, no common utility level can be computed for them. Should the relocation systematically reduce the utility of public-sector workers, the aggregate welfare effects of the relocation could turn negative.

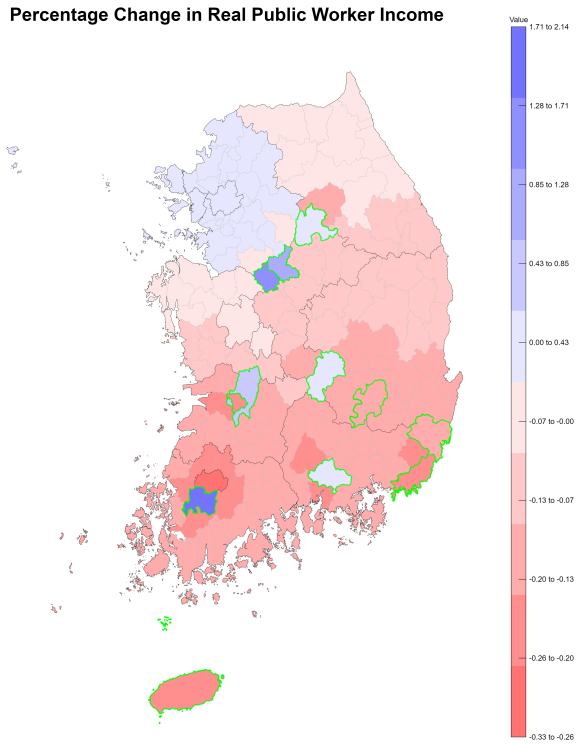
Although utility for public-sector workers is not directly observable, changes in their real wages $\tilde{v}_n^G$ serve as an indicator of the direction of welfare effects:

$$\tilde{v}_n^G = \frac{v_n^G}{{P_{nT}}^{\alpha_T} {P_{nN}}^{\alpha_N} {Q_n}^{1-\alpha_T-\alpha_N}}, \tag{29}$$

where v_n^G gives residential after-tax income in the public sector and the denominator comprises the price indices for tradable goods, non-tradable goods, and housing.

Figure 5 shows changes in real wages for public-sector workers. Real wages rise marginally in the Seoul Metropolitan Area but decline in districts outside it. Workers remaining in the Seoul Metropolitan Area benefit from lower prices for non-tradable services, reflecting the decline in private-sector wages there; these lower wages, combined with reduced housing competition and a larger local tradable sector, also lower the prices of tradable goods and housing in the Seoul Metropolitan Area, with the inverse holding in locations outside it. Effects across the Innovation

Figure 5: Public Real Wage Changes of Counterfactual Innovation City Project



Notes: Percentage Changes in county-level public-sector worker real wages post counterfactual Innovation City Project, including public-sector relocation and housing construction. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red.

Cities are mixed: western Innovation Cities record increases in public-sector real wages, while eastern Innovation Cities record declines. With the exception of Gimcheon, all western Innovation City districts have large shares of in-commuting workers; combined with the accompanying housing construction, these districts therefore see the largest declines in housing costs, raising real incomes. Gimcheon records a particularly strong outflow of tradable-sector employment, which also drives down housing costs. The policy is therefore likely welfare-enhancing for public-sector workers remaining in the Seoul Metropolitan Area, while reducing the welfare of those originally located outside it, with the exception of workers in the western Innovation Cities. For relocated workers, the welfare effects are ambiguous: since the public-sector wage is uniform across locations and price levels in the Seoul Metropolitan Area are considerably higher, relocated workers likely move to a location where their income affords greater consumption. The aggregate real income of the entire public sector increases marginally by 0.3 %, though this gain must be weighed against potential amenity losses from leaving the Seoul Metropolitan Area and, for workers who do not relocate their residence, the disutility of longer commutes.

7.3 Isolated Relocation

The model-implied employment changes confirm the reduced-form evidence. Although the reduced-form estimates show more positive effects at the Innovation Cities themselves, both approaches identify the same underlying mechanism: limited negative or null effects on manufacturing employment and private employment gains driven by the service sector. The model likewise finds no meaningful spillovers to surrounding regions in either a positive or negative direction.

The stronger reduced-form effects most likely reflect the exclusion of auxiliary aspects of the Innovation City Project, as well as other concurrent regional policies. Besides the residential construction, the Innovation City Project included amenity construction, such as sports and cultural facilities, which raised the attractiveness

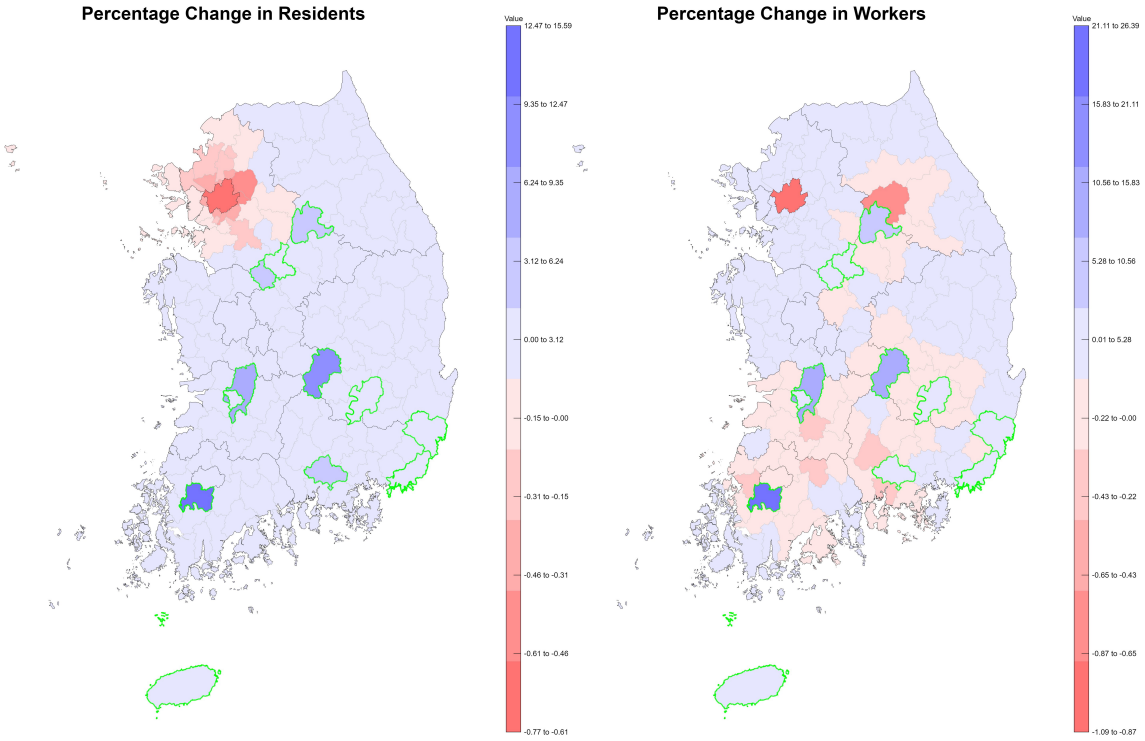
of the Innovation Cities and thereby amplified inflows of both residents and jobs.

Apart from the policy itself multiple other concurrent regional policies might confound the results as well. A full overview of related policies is given in Section D, which can be roughly divided into two segments. A range of laws, most notably the "Capital Region Readjustment Planning Act", restrict new construction in the Seoul Metropolitan Area, with some additional tax laws further penalizing economic activity compared to other provinces. These laws would prevent or limit the outflow of tradable employment back to the Seoul Area, given the increased cost of setting up businesses there. With Seoul relatively less attractive, more tradable employment might stay in the Innovation Cities. The second group of place-based policies contains interventions at the Innovation Cities themselves. Most counties receiving Innovation City status also got targeted by other interventions in parallel, such as the construction of industrial complexes or improved connectivity within the South Korean high-speed rail network. These other changes might increase the attractiveness of the targeted counties independently of the Innovation City Project. Including and quantifying these other policies is left to future research.

Instead of adding further policy dimensions to the baseline, the second counterfactual isolates the public-sector relocation from all accompanying interventions. The baseline counterfactual combines the relocation with housing construction, reflecting the full policy package in which additional investments support the relocation's impact. The isolated counterfactual strips these away to identify the effect of the relocation itself. This provides evidence on the effectiveness of the underlying policy instrument independently of the auxiliary interventions that accompanied it.

Figure 6 shows the effects of the isolated counterfactual on residents and total employment. The general trends of the baseline counterfactual persist, but with weaker effect sizes. The outflow of residents stems exclusively from the Seoul Metropolitan Area and declines to around 42,000; of these relocating households, only 78 % move to the Innovation Cities, as opposed to 90 % in the baseline counterfactual. Since

Figure 6: Resident and Worker Changes of Isolated Public-Sector Relocation

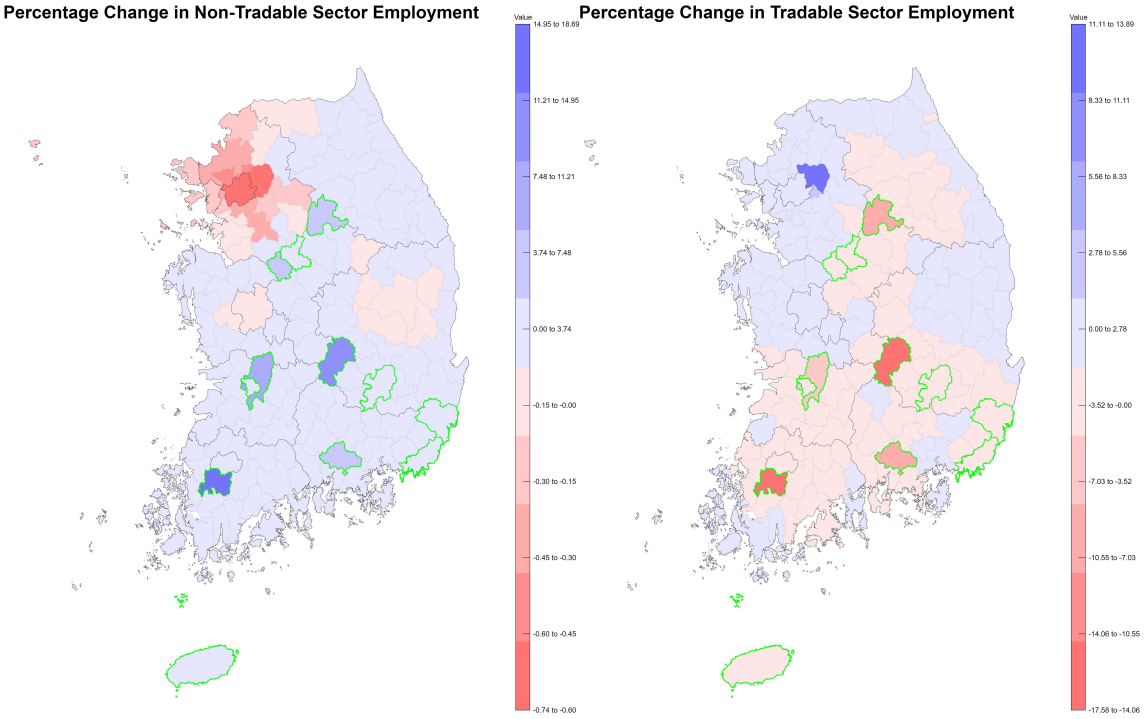


Notes: Percentage Changes in county-level residents and workers post isolated public-sector relocation. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red. Left residents, right total employment.

no additional housing is constructed, more relocated workers choose residences outside their new workplace and commute. The employment outflow from the Seoul Metropolitan Area similarly declines in magnitude, with only 43,000 workers departing, again entirely to the Innovation Cities. The pattern of employment losses in counties neighboring the Innovation Cities also persists and widens marginally, affecting more surrounding counties, while those counties further away no longer experience an outflow of workers.

Decomposing the private sector into its two components (see Figure 7) reveals the directional effects of the relocation alone. Contrary to the baseline counterfactual, all Innovation Cities gain non-tradable employment at the expense of tradable employment. Non-tradable employment declines almost exclusively in the Seoul Metropolitan Area, where it is counterbalanced by an inflow of tradable employment. The tradable sector declines across all Innovation Cities, losing over 16,000

Figure 7: Private Non-Tradable and Tradable Employment Changes of Isolated Public-Sector Relocation



Notes: Percentage Changes in county-level residents and workers post isolated public-sector relocation. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red. Left non-tradable employment, right tradable employment.

workers in total. Surrounding locations lose a further 5,900 tradable jobs, while remaining locations outside the Seoul Metropolitan Area gain 700. With no housing construction at the Innovation Cities, price pressures there are stronger, displacing more workers into surrounding counties and reducing tradable employment in both. The more limited population outflow also produces a smaller increase in non-tradable employment, and the public-sector multipliers decline accordingly: each relocated public-sector job creates 0.43 non-tradable jobs and crowds out 0.39 tradable jobs.

The isolated counterfactual multipliers are in the range of those reported by Faggio and Overman (2014) for a public-sector expansion in the United Kingdom at the local authority level, an administrative unit comparable to the South Korean counties examined here. Faggio and Overman (2014) find an employment multiplier of public-sector workers on the service sector of 0.5 and of -0.4 on the manufacturing

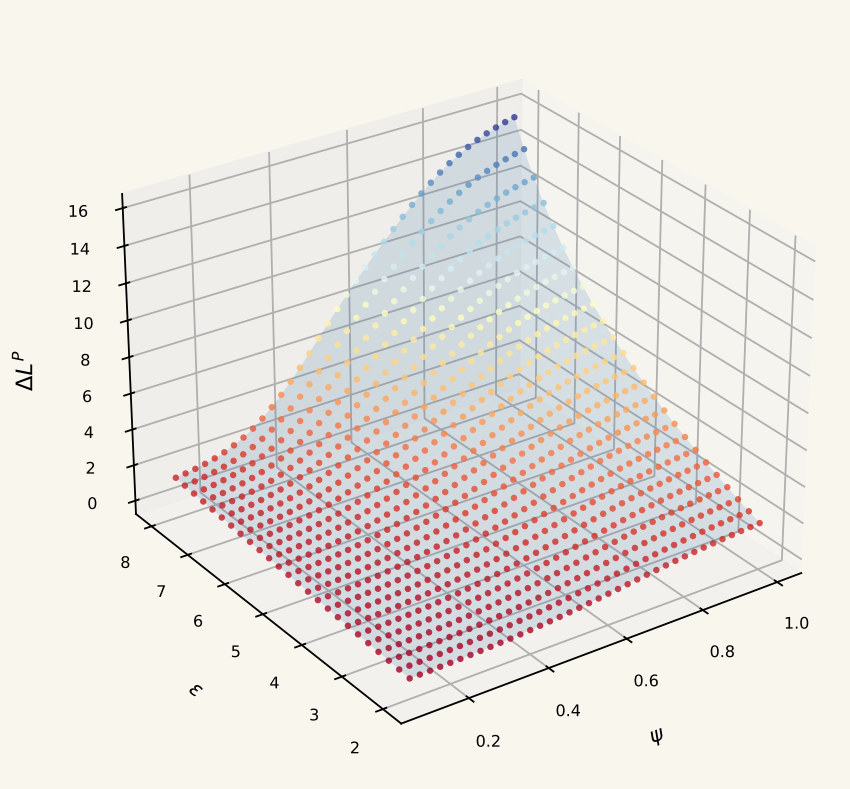
sector. The nature of the shock examined by Faggio and Overman (2014), a general expansion of the public sector, also aligns closely with the isolated relocation counterfactual, since in both cases the targeted regions are affected solely by an increase in public-sector employment without accompanying investments or interventions. Despite the difference in country context, the similarity in results suggests that the model accurately replicates the employment effects of the public-sector relocation itself.

7.4 Robustness

In this final section the robustness of the prior results is examined, beginning with the trade elasticity. Figure 8 shows the change in private employment in the Innovation Cities for different combinations of the shape parameter $\varepsilon \in [2, 8]$ and the trade cost elasticity $\psi \in [0.1, 1]$. The change in private employment increases in both ε and ψ. These increases are to be expected, as a higher trade cost elasticity makes it more costly to ship tradable goods over large distances, reducing the outflow of the tradable sector from the Innovation Cities. Using a value of $\psi = 0.43$, as found by Monte, Redding, and Rossi-Hansberg (2018), would increase the change in private employment by 40 %, mainly driven by lower outflows of tradable sector workers. A higher shape parameter represents more homogeneous households, making them more sensitive to local prices and wages, amplifying the response to the improved conditions at the targeted relocation counties.

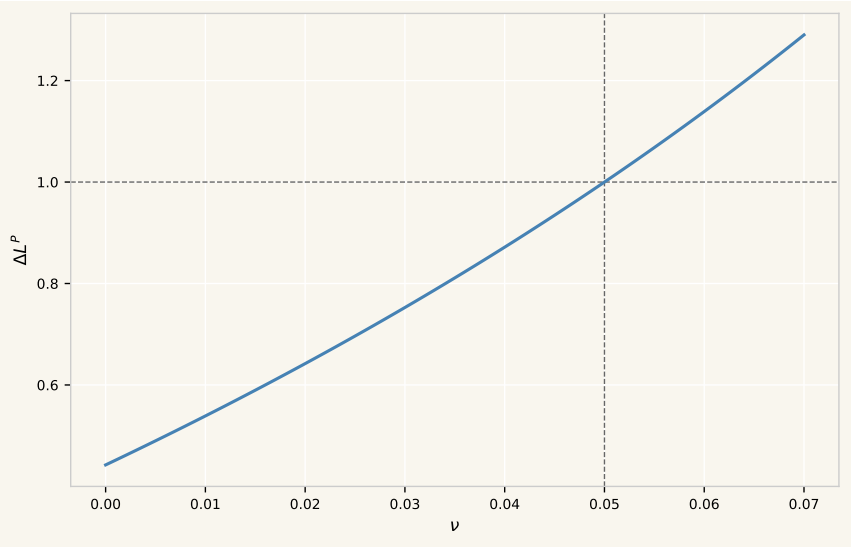
As a second robustness test the agglomeration elasticity ν is varied. The results of this test are shown in Figure 9. Without agglomeration economies, defined by $\nu = 0$, the effect of the Innovation City Project is cut in half. Since public-sector workers provide no utility for local residents, they can only improve a location's attractiveness through increased productivity. Without the productivity channel, public-sector workers still induce some employment effects, due to increased local demand. But the resulting local wage increases lead to a large crowding out of the

Figure 8: Robustness with Respect to ε and ψ



Notes: Figure shows change in private employment in Innovation cities for alternative values of shape parameter ε (Y-axis) and trade cost elasticity ψ (X-axis). Change in private employment indexed to 1 for baseline setting of $\varepsilon = 4.01$ and $\psi = 0.31$.

Figure 9: Robustness with Respect to ν



Notes: Figure shows change in private employment in Innovation cities for alternative values of agglomeration elasticity ν. Change in private employment indexed to 1 for baseline setting of $\nu = 0.05$.

private sector, which no longer benefits from the increased productivity.

8 Conclusion

This paper develops a quantitative spatial model to examine the effects of isolated public-sector relocations on local employment, differentiating between a tradable and a non-tradable private sector. Calibrating the model for South Korea and the Innovation City Project, the estimates show that public-sector relocation combined with housing construction increases non-tradable employment by 0.62 workers per relocated public-sector worker, while each relocated public-sector worker simultaneously crowds out 0.24 workers in the tradable sector. The private employment gains come largely at the expense of the Seoul Metropolitan Area, distinguishing the Innovation City Project from conventional place-based policies such as industrial complexes or economic zones, where employment gains at the targeted location are typically drawn from activity in surrounding regions.

The modeled Innovation City Project fulfills the core goal of the policy, moving

economic activity from the Seoul Metropolitan Area to the Innovation Cities. The policy also produces very limited spillovers: on net, 90 % of relocated residents move to the Innovation Cities, and the entirety of net employment losses in the Seoul Metropolitan Area are absorbed by the Innovation Cities. Private employment in the immediate surroundings of the Innovation Cities declines marginally, with only $1,500$ workers displaced to either more distant counties or into the Innovation Cities. These localized changes leave overall welfare intact, producing negligible but positive private-sector utility gains and public-sector real wage increases, while redistributing activity out of the Seoul Metropolitan Area. The policy falls short, however, of its secondary goals of raising local productivity and attracting industry. Although local productivity rises on account of increased local employment, the gains are insufficient to offset the higher production costs arising from increased wages, which crowd out tradable employment contrary to policy goals. Private employment gains instead stem from the non-tradable sector, driven by increased local demand.

The structural multiplier of 0.38 falls below the $1.0 - 1.4$ range observed in the reduced-form evidence and literature, and excluding housing construction reduces it further to 0.04. The divergence highlights the importance of auxiliary policy interventions in enabling public-sector relocations to generate meaningful private employment gains. Quantifying the contribution of such interventions beyond housing construction, including infrastructure investment, amenity improvements, and restrictive zoning laws, is left to future research.

References

Abramovsky, Laura, Rupert Harrison, and Helen Simpson. 2007. "University Research and the Location of Business R&D." *The Economic Journal* 117 (519): C114–C141. https://doi.org/10.1111/j.1468-0297.2007.02038.x.

Andersson, Roland, John M Quigley, and Mats Wilhelmsson. 2009. "Urbanization, Productivity, and Innovation: Evidence from Investment in Higher Education." *Journal of Urban Economics* 66 (1): 2–15. https://doi.org/10.1016/j.jue.2009.02.004.

Bang, Seolah. 2025. "Construction of the Multifunctional Administrative City." *A Primer on Korean Planning and Policy,* nos. 2025-02, accessed May 13, 2026. https://www.krihs.re.kr/gallery.es?act=view&bid=0026&list_no=32346&mid=a20203000000.

Bartik, Timothy J., and Nathan Sotherland. 2019. "Local Job Multipliers in the United States: Variation with Local Characteristics and with High-Tech Shocks." *Upjohn Institute Working Paper,* nos. 19-301, https://doi.org/10.17848/wp19-301.

Bayerisches Staatsministerium der Finanzen, für Landesentwicklung und Heimat (StMFH). 2014. "Heimatstrategie: Starke Zukunft für Stadt und Land." (München), accessed May 13, 2026. https://www.stmfh.bayern.de/heimat/heimatstrategie.pdf.

Becker, Sascha O, Peter H Egger, and Maximilian Von Ehrlich. 2010. "Going NUTS: The Effect of EU Structural Funds on Regional Performance." *Journal of Public Economics* 94 (9-10): 578–590. https://doi.org/10.1016/j.jpubeco.2010.06.006.

Becker, Sascha O, Stephan Heblich, and Daniel M Sturm. 2021. "The Impact of Public Employment: Evidence from Bonn." *Journal of Urban Economics* 122. https://doi.org/10.1016/j.jue.2020.103291.

Blouri, Yashar, and Maximilian Von Ehrlich. 2020. "On the Optimal Design of Place-Based Policies: A Structural Evaluation of EU regional Transfers." *Journal of International Economics* 125. https://doi.org/10.1016/j.jinteco.2020.103319.

Briant, Anthony, Miren Lafourcade, and Benoît Schmutz. 2015. "Can Tax Breaks Beat Geography? Lessons from the French Enterprise Zone Experience." *American Economic Journal: Economic Policy* 7 (2): 88–124. https://doi.org/10.1257/pol.20120137.

Broda, Christian, and David E. Weinstein. 2006. "Globalization and the Gains from Variety." *The Quarterly Journal of Economics* 121 (2): 541–585. https://doi.org/10.1162/qjec.2006.121.2.541.

Busso, Matias, Jesse Gregory, and Patrick Kline. 2013. "Assessing the Incidence and Efficiency of a Prominent Place Based Policy." *American Economic Review* 103 (2): 897–947. https://doi.org/10.1257/aer.103.2.897.

Combes, Pierre-Philippe, Gilles Duranton, Laurent Gobillon, Diego Puga, and Sébastien Roux. 2012. "The Productivity Advantages of Large Cities: Distinguishing Agglomeration from Firm Selection." *Econometrica* 80 (6): 2543–2594. https://doi.org/10.3982/ECTA8442.

De Blasio, Guido, and Carlo Menon. 2011. "Local Effects of Manufacturing Employment Growth in Italy." *Giornale degli Economisti e Annali di Economia,* 101–112. http://www.jstor.org/stable/41756381.

Faggio, Giulia. 2019. "Relocation of Public Sector Workers: Evaluating a Place-Based Policy." *Journal of Urban Economics* 111:53–75. https://doi.org/10.1016/j.jue.2019.03.001.

Faggio, Giulia, and Henry Overman. 2014. "The Effect of Public Sector Employment on Local Labour Markets." *Journal of Urban Economics* 79:91–107. https://doi.org/10.1016/j.jue.2013.05.002.

Fajgelbaum, Pablo D, Eduardo Morales, Juan Carlos Suárez Serrato, and Owen Zidar. 2019. "State Taxes and Spatial Misallocation." *The Review of Economic Studies* 86 (1): 333–376. https://doi.org/10.1093/restud/rdy050.

Finansministeriet. 2015. *Bedre Balance: Statslige Arbejdspladser Tættere På Borgere og Virksomheder*. København: Finansministeriet. Accessed May 13, 2026. https://gst.dk/media/6682/bedre-balance-_-statslige-arbejdspladser-taettere-paa-borgere-og-virksomheder_web-1.pdf.

Freedman, Matthew, Shantanu Khanna, and David Neumark. 2023. "The Impacts of Opportunity Zones on Zone Residents." *Journal of Urban Economics* 133. https://doi.org/10.1016/j.jue.2021.103407.

Freitas, Dimitria. 2025. "The Effect of Public Sector Relocations on Regional Development in Germany." In *VfS Annual Conference 2025: Revival of Industrial Policy*. Verein für Socialpolitik. Accessed May 13, 2026. https://www.econstor.eu/handle/10419/325457.

Gobillon, Laurent, Thierry Magnac, and Harris Selod. 2012. "Do Unemployed Workers Benefit from Enterprise Zones? The French Experience." *Journal of Public Economics* 96 (9-10): 881–892. https://doi.org/10.1016/j.jpubeco.2012.06.003.

Guillouzouic, Arthur, Emeric Henry, and Joan Monras. 2024. "Public Sector Pay in Spatial Equilibrium." *CEPR Discussion Papers,* no. 19014, accessed May 13, 2026. https://cepr.org/publications/dp19014.

Head, Keith, and Thierry Mayer. 2014. "Gravity Equations: Workhorse, Toolkit, and Cookbook." In *Handbook of International Economics,* 4:131–195. Elsevier. https://doi.org/10.1016/B978-0-444-54314-1.00003-3.

Henkel, Marcel, Tobias Seidel, and Jens Suedekum. 2021. "Fiscal Transfers in the Spatial Economy." *American Economic Journal: Economic Policy* 13 (4): 433–468. https://doi.org/10.1257/pol.20180294.

Jaffe, Adam B. 1989. "Real Effects of Academic Research." *The American Economic Review* 79 (5): 957–970. http://www.jstor.org/stable/1831431.

Jensen, Peter Sandholt, Torben Dall Schmidt, and Jens Fyhn Lykke Sørensen. 2025. "Relocating to Peripheral Cities: Development and Diversity From Relocating Government Jobs." *Journal of Regional Science,* https://doi.org/10.1111/jors.70040.

Jofre-Monseny, Jordi, José I Silva, and Javier Vázquez-Grenno. 2020. "Local Labor Market Effects of Public Employment." *Regional Science and Urban Economics* 82:103406. https://doi.org/10.1016/j.regsciurbeco.2018.11.001.

Kang, Song Hee, Jae Seung Lee, and Saehoon Kim. 2024. "Has South Korea's Policy of Relocating Public Institutions been Successful? A Case Study of 12 Agglomeration Areas under the Innovation City Policy." *Urban Studies* 61 (5): 900–922. https://doi.org/10.1177/00420980231193567.

Kim, Gwang-ik, Dong-woo Lee, Young-seop Kwon, Hyeong-jin Kim, and Gyeonghyeon Park. 2006. "A Study on the White Paper for the Public Agency Relocation in Capital Region and the Inno City Construction [공공기관 지방이전 및 혁신도시 건설백서 연구]." *KRIHS Research Material 2006-13* (Anyang, Republic of Korea), accessed May 13, 2026. https://library.krihs.re.kr/%24/10110/contents/6158342.

Kim, Woo-Yung, and Sung Hyo Hong. 2019. "Local Employment Multipliers when Living and Working Areas are Different." *Economics Letters* 175:47–50. https://doi.org/10.1016/j.econlet.2018.12.028.

Kline, Patrick, and Enrico Moretti. 2014. "Local Economic Development, Agglomeration Economies, and the Big Push: 100 Years of Evidence from the Tennessee Valley Authority." *Quarterly Journal of Economics* 129 (1): 275–331. https://doi.org/10.1093/qje/qjt034.

Korea Research Institute for Human Settlements (KRIHS). 2022a. "The Fifth Comprehensive National Territorial Plan (2020–2040)." *Primer on Korean Planning and Policy,* nos. 2022-05, accessed May 20, 2026. https://www.krihs.re.kr/gallery.es?mid=a20203000000&bid=0026&list_no=30168&act=view.

———. 2022b. "The Fourth Comprehensive National Territorial Plan (2000–2020)." *Primer on Korean Planning and Policy,* nos. 2022-04, accessed May 20, 2026. https://www.krihs.re.kr/gallery.es?mid=a20203000000&bid=0026&list_no=30167&act=view.

Korea Research Institute for Human Settlements (KRIHS). 2022c. "The Second Comprehensive National Territorial Plan (1982–1991)." *Primer on Korean Planning and Policy,* nos. 2022-02, accessed May 20, 2026. https://www.krihs.re.kr/gallery.es?mid=a20203000000&bid=0026&act=view&list_no=30165.

———. 2022d. "The Third Comprehensive National Territorial Plan (1992–2001)." *Primer on Korean Planning and Policy,* nos. 2022-03, accessed May 20, 2026. https://www.krihs.re.kr/gallery.es?mid=a20203000000&bid=0026&tag=&b_list=10&act=view&list_no=30166&nPage=1&vlist_no_npage=0&keyField=&keyWord=&orderby=.

Korea Transport Institute (KTI). 2011. "2010 National Interregional Freight Origin-Destination Update(2010[년)국가교통수요조사 및 DB구축사업.]," accessed May 13, 2026. https://dl.nanet.go.kr/detail/MONO1201135195.

Krugman, Paul. 1980. "Scale Economies, Product Differentiation, and the Pattern of Trade." *American Economic Review* 70 (5): 950–959. http://www.jstor.org/stable/1805774.

Lee, Hwanoong, Changsu Ko, and Wookun Kim. 2024. "Local Employment Multiplier: Evidence from Relocation of Public-Sector Entities in South Korea." *Labour Economics* 88. https://doi.org/10.1016/j.labeco.2024.102515.

Lee, Jae Kwang. 2021. "Transport Infrastructure Investment, Accessibility Change and Firm Productivity: Evidence from the Seoul Region." *Journal of Transport Geography* 96. https://doi.org/10.1016/j.jtrangeo.2021.103182.

Lee, Jinwoong. 2024. "Changes in the Price Elasticity of New Housing Supply in Korea: A Panel Data Analysis by Province and Metropolitan City[한국의 신규 주택공급 가격탄력성 변화: 시도별 패널데이터 분석]." *Journal of International Economic Studies* 30 (3): 1–28. https://doi.org/10.17298/kky.2024.30.3.001.

Lee, Sang Keon, Heeyoun You, and Heeseo Rain Kwon. 2015. "Korea's Pursuit for Sustainable Cities through New Town Development: Implications for LAC," https://doi.org/10.18235/0006999.

Lu, Yi, Jin Wang, and Lianming Zhu. 2019. "Place-Based Policies, Creation, and Agglomeration Economies: Evidence from China's Economic Zone Program." *American Economic Journal: Economic Policy* 11 (3): 325–360. https://doi.org/10.1257/pol.20160272.

Lyons, Sir Michael. 2004. *Well Placed to Deliver?: Shaping the Pattern of Government Service: Independent Review of Public Sector Relocation.* HM Stationery Office. Accessed May 13, 2026. https://www.civilservant.org.uk/library/2004-Lyons-full_report.pdf.

Ministry of Land, Infrastructure and Transport (MOLIT). 2026. *Innovation City Portal/*혁신도시 종합정보시스템]. Republic of Korea, Ministry of Land, Infrastructure and Transport. Accessed May 13, 2026. https://innocity.molit.go.kr/index.do.

Monte, Ferdinando, Stephen J Redding, and Esteban Rossi-Hansberg. 2018. "Commuting, Migration, and Local Employment Elasticities." *American Economic Review* 108 (12): 3855–3890. https://doi.org/10.1257/aer.20151507.

Moretti, Enrico. 2010. "Local Multipliers." *American Economic Review* 100 (2): 373–377. https://doi.org/10.1257/aer.100.2.373.

Moretti, Enrico, and Per Thulin. 2013. "Local Multipliers and Human Capital in the United States and Sweden." *Industrial and Corporate Change* 22 (1): 339–362. https://doi.org/10.1093/icc/dts051.

Neumark, David, and Helen Simpson. 2015. "Place-Based Policies." In *Handbook of Regional and Urban Economics,* edited by Gilles Duranton, J. Vernon Henderson, and William C. Strange, 5:1197–1287. Elsevier. https://doi.org/10.1016/B978-0-444-59531-7.00018-1.

OECD. 2012. *Industrial Policy and Territorial Development: Lessons from Korea.* Development Centre Studies. OECD Publishing. https://doi.org/10.1787/9789264173897-en.

———. 2026. *National CPI, Item weights – Korea.* OECD Data Explorer, OECD.Stat. Accessed May 13, 2026. https://data-explorer.oecd.org/vis?df[ds]=dsDisseminateFinalDMZ&df[id]=DSD_PRICES%40DF_PRICES_N_IT_W&df[ag]=OECD.SDD.TPS&dq=KOR.A.....&pd=2010%5C%2C2010.

Osman, Taner, and Tom Kemeny. 2022. "Local Job Multipliers Revisited." *Journal of Regional Science* 62 (1): 150–170. https://doi.org/10.1111/jors.12561.

Park, Kyunghyun. 2024. "Regional Development for Balanced National Development." *Primer on Korean Planning and Policy,* nos. 2024-02, accessed May 13, 2026. https://www.krihs.re.kr/gallery.es?mid=a20203000000&bid=0026&act=view&list_no=32153.

Presidential Advisory Policy Planning Committee (PAPPC). 2008. "Relocation of Public Institutions and Construction of Innovation Cities [공공기관 지방이전 및 혁신도시 건설]." *Government Policy Report,* Participatory Government Policy Reports [참여정부 정책보고서], nos. 2-05, accessed May 13, 2026. https://archives.knowhow.or.kr/policy/report/view/16532.

Rosenthal, Stuart S, and William C Strange. 2004. "Evidence on the Nature and Sources of Agglomeration Economies." In *Handbook of Regional and Urban Economics,* 4:2119–2171. Elsevier. https://doi.org/10.1016/S1574-0080(04)80006-3.

Seidel, Tobias, and Jan Wickerath. 2020. “Rush Hours and Urbanization.” *Regional Science and Urban Economics* 85. https://doi.org/10.1016/j.regsciurbeco.2020.103580.

Seo, Seongmin, and Won Kwak Do. 2023. “The Impact of the Innovation City Project on the Local Economy: Evidence from Korean Innovation City Project from 2012 to 2014.” *Journal of Asian Economics,* 101677. https://doi.org/10.1016/j.asieco.2023.101677.

Song, Woo Kyung, Man Tae Jeong, Jeong Seok Ha, and Seong Min Jo. 2020. “Current Status and Development Directions of Enterprise Cities [기업도시의 추진현황과 발전방향]” [in Korean]. *Issue Paper* (Seoul), nos. 2020-15, accessed May 13, 2026. https://nkis.re.kr/subject_view1.do?otpId=OTP_0000000000005323&otpSeq=0.

Soo, Kwok Tong. 2005. “Zipf’s Law for Cities: a Cross-Country Investigation.” *Regional Science and Urban Economics* 35 (3): 239–263. https://doi.org/10.1016/j.regsciurbeco.2004.04.004.

United States Forces Korea (USFK). 2021. *Guide to Safe Driving in Korea.* USFK Manual 4800.02. United States Forces Korea. Accessed May 13, 2026. https://www.usfk.mil/Portals/105/Documents/Publications/Pamplets/USFK-Pam-385-2-Guide-to-Safe-Driving-in-Korea-20200521.pdf.

Veneri, Paolo. 2013. “On City Size Distribution: Evidence from OECD Functional Urban Areas.” *OECD Regional Development Working Papers* 2013 (27). https://doi.org/10.1787/5k3tt100wf7j-en.

Wang, Jin. 2013. “The Economic Impact of Special Economic Zones: Evidence from Chinese Municipalities.” *Journal of development economics* 101:133–147. https://doi.org/10.1016/j.jdeveco.2012.10.009.

Yoon, Youngmo. 2019. *Establishing a Foundation for Innovation-Creating Regional Development: Focusing on Innovation Cities[*혁신창출형 지역발전 기반 구축 연구: 혁신도시를 중심으로]. Technical report. Korea Research Institute for Human Settlements. Accessed May 22, 2026. https://www.krihs.re.kr/gallery.es?mid=a10605000000&bid=0029&act=view&list_no=29375.

Table A.1: Structural Parameters

Parameter	**Function**	**Value**
α_T	Tradable expenditure share.	0.39
α_N	Non-tradable expenditure share.	0.44
$1 - \alpha_T - \alpha_N$	Housing expenditure share.	0.17
σ	Elasticity of substitution across tradable varieties.	4
F	Fixed labor requirement per tradable variety.	1
ψ	Freight trade-cost elasticity.	0.31
ϵ	Preference heterogeneity.	4.01
γ	Commuting-cost elasticity.	1.00
δ	Housing supply elasticity.	1.71
ν	Agglomeration elasticity.	0.05

Appendix

A Algorithms

Algorithm 1: Computes public budgets, spending shares, and the uniform tax rate: `compute_tax.m`

Data: Private employment L_n^P, public employment L_n^G, public-sector wages w_n^G, and private-sector wages w_n^P.

1 Compute local public wage bills:

$$G_n = w_n^G L_n^G.$$

2 Compute total public budget:

$$G = \sum_n G_n.$$

3 Compute the aggregate wage tax base:

$$\mathcal{T} = \sum_n w_n^P L_n^P + \sum_n w_n^G L_n^G.$$

4 Compute the uniform tax rate:

$$t = \frac{G}{\mathcal{T}}.$$

5 Compute local public spending shares:

$$g_n = \frac{G_n}{G}.$$

Result: Local public budgets G_n, total public budget G, local spending shares g_n, and uniform tax rate t.

Algorithm 2: Solving tradable productivity, trade shares, and price indices: `solveProductTradeTK_new.m`

Data: Private tradable workplace employment L^P_{nT}, workplace private wages w^P_n, local income ${Y^{loc}}_n$, bilateral freight travel times d^T_{ni}, and structural parameters $\{\sigma, F, \alpha_T, \alpha_N, \psi\}$.

Initialize: Set productivity guesses $A^0_i = 1$ for all locations i. Choose maximum iterations $\bar{m}$, convergence precision, and convex combination parameter ξ.

while $m < \bar{m}$ **do**

Compute bilateral trade attractiveness:

$$q^m_{ni} = {d^T_{ni}}^{\psi(1-\sigma)} (A^m_i)^{\sigma-1} L^P_{iT} (w^P_i)^{1-\sigma}.$$

Compute bilateral trade shares:

$$\pi^m_{ni} = \frac{q^m_{ni}}{\sum_k q^m_{nk}}.$$

Compute workplace income and tradable expenditure paid to each workplace location:

$$Y^m_i = w^P_i L^P_{iT}, \qquad X^m_i = \frac{\alpha_T}{\alpha_T + \alpha_N} \sum_n \pi^m_{ni} {Y^{loc}}_n.$$

Compute convergence gap:

$$\Delta^m_i = \left| Y^m_i - X^m_i \right|.$$

Algorithm continues on the next page.

Algorithm 2: Solving tradable productivity, trade shares, and price indices (continued)

if *all rounded gaps satisfy* $\Delta_i^m = 0$ **then**

Recompute final trade shares π_{ni} using converged productivity A_i.

Compute own trade shares:

$$\pi_{ii} = \text{diag}(\pi_{ni}).$$

Extract internal trade costs:

$$d_{ii}^T = \text{diag}(d_{ni}^T).$$

Compute tradable price index:

$$P_i = \frac{\sigma}{\sigma - 1} \left(\frac{L_{iT}^P}{\sigma F \pi_{ii}} \right)^{\frac{1}{1-\sigma}} \frac{{d_{ii}^T}^{\psi} w_i^P}{A_i}.$$

Result: Productivity A_i, bilateral trade shares π_{ni}, own trade shares π_{ii}, and tradable price index P_i.

return

Update productivity guesses:

$$\widetilde{A}_i^{m+1} = A_i^m \frac{Y_i^m}{X_i^m}.$$

Apply convex updating:

$$A_i^{m+1} = \xi \widetilde{A}_i^{m+1} + (1 - \xi) A_i^m.$$

Set $m \leftarrow m + 1$.

If convergence is not achieved within $\bar{m}$ iterations, stop and report non-convergence.

Algorithm 3: Solving bilateral amenities: `solveAmen_alt.m`

Data: Private workplace employment L^P_{nT}, available land $\bar{H}_n$, private workplace wages w^P_n, local income ${Y^{loc}}_n$, bilateral commuting travel time d_{ni}, bilateral freight travel time d^T_{ni}, productivities A_n, own trade shares π_{nn}, observed unconditional commuting probabilities λ_{ni}, and structural parameters $\{\sigma, \alpha_T, \alpha_N, \delta, \epsilon, \psi, \gamma\}$.

Initialize: Set bilateral amenity guesses $B^0_{ni} = 1$ for all origin–destination pairs (n, i). Extract internal freight travel time $d^T_{ii} = \text{diag}(d^T_{ni})$. Choose maximum iterations $\bar{m}$, convergence precision, and convex combination parameter ξ.

while $m < \bar{m}$ **do**

Compute the non-wage component of bilateral commuting attractiveness:

$$Q^m_{ni} = B^m_{ni} \left(\frac{L^P_{nT}}{\pi_{nn}} \right)^{-\frac{\alpha_T \epsilon}{1-\sigma}} \left(\frac{{d^T_{nn}}^{\psi} w^P_n}{A_n} \right)^{-\alpha_T \epsilon} \left(\frac{w^P_n}{A_n} \right)^{-\alpha_N \epsilon} \left(\frac{{Y^{loc}}_n}{\bar{H}_n} \right)^{-\frac{(1-\alpha_N-\alpha_T)\epsilon}{1+\delta}} d_{ni}^{-\gamma\epsilon}.$$

Compute bilateral commuting attractiveness:

$$\widetilde{Q}^m_{ni} = {w^P_i}^{\epsilon} Q^m_{ni}.$$

Compute implied unconditional commuting probabilities:

$$\widehat{\lambda}^m_{ni} = \frac{\widetilde{Q}^m_{ni}}{\sum_r \sum_s \widetilde{Q}^m_{rs}}.$$

Compute convergence gap:

$$\Delta^m_{ni} = \left| \lambda_{ni} - \widehat{\lambda}^m_{ni} \right|.$$

if *all rounded gaps satisfy* $\Delta^m_{ni} = 0$ **then**

Result: Bilateral amenity matrix B_{ni}.

return

Update bilateral amenity guesses:

$$\widetilde{B}^{m+1}_{ni} = B^m_{ni} \frac{\lambda_{ni}}{\widehat{\lambda}^m_{ni}}.$$

Apply convex updating:

$$B^{m+1}_{ni} = \xi \widetilde{B}^{m+1}_{ni} + (1 - \xi) B^m_{ni}.$$

Set $m \leftarrow m + 1$.

If convergence is not achieved within $\bar{m}$ iterations, stop and report non-convergence.

Algorithm 4: Constructing counterfactual land supply and added income: `make_alt_land.m`

Data: Local income ${Y^{loc}}_n$, residents R_n, additionally constructed housing units Units_n, housing prices Q_n, and structural parameters $\{\alpha_T, \alpha_N, \delta\}$.

34 Compute average local income per resident:

$$v_n = \frac{{Y^{loc}}_n}{R_n}.$$

35 Compute additional local income assuming newly constructed units are fully filled at current income:

$$\text{AddInc}_n = v_n \text{Units}_n.$$

36 Compute counterfactual local income:

$${Y^{loc}}^{cf}_n = {Y^{loc}}_n + \text{AddInc}_n.$$

37 Compute counterfactual land supply:

$$\bar{H}_n = (1 - \alpha_T - \alpha_N)\frac{{Y^{loc}}^{cf}_n}{Q_n^{1+\delta}}.$$

Result: Counterfactual land supply $\bar{H}_n$ and added income AddInc_n.

Algorithm 5: Counterfactual equilibrium with public employment: `LamW_CF_full_V2.m`

Data: Initial private wages w_n^P, initial unconditional private commuting probabilities λ_{ni}^P, commuting travel time d_{ni}, freight travel time d_{ni}^T, bilateral amenities B_{ni}, total private employment L^P,altered public employment L_n^G, public wages w_n^G, land supply $\bar{H}_n$, baseline productivity shifters $\bar{A}_n$, and structural parameters $\{\nu, \delta, \alpha_N, \alpha_T, \sigma, \epsilon, \gamma, \psi\}$.

Initialize: Set wage guesses $w_n^0 = w_n^P$ and commuting-probability guesses $\lambda_{ni}^0 = \lambda_{ni}^P$. Set convergence status to false and initialize failure diagnostics.

Run the prerun adjustment routine (see Algorithm 6) until five consecutive rounds produce nonnegative tradable employment L_{nT}^P

if *the prerun adjustment fails* **then**

  Return current guesses and failure diagnostics

Set maximum iterations $\bar{m}$ and iteration counter $m = 0$

while true do

  Set $m \leftarrow m + 1$

  Compute conditional workplace probabilities:

$$\lambda_{ni}^W = \text{updateCondCom}(\lambda_{ni}^0).$$

  Compute private workplace employment:

$$L_n^P = \text{updateEmpl}(\lambda_{ni}^0, L^P).$$

  Compute the uniform tax rate:

$$t = \text{updateTax}(w_n^G, w_n^0, L_n^P, L_n^G).$$

  Compute expected private and public residential wages:

$$v_n^P = \text{updateResWage_priv_alt}(w_n^0, d_{ni}, t, B_{ni}), \qquad v_n^G = \text{updateResWage_pub}(w_n^G, t).$$

  Compute private and public residents:

$$R_n^P = \text{updateRes_priv}(\lambda_{ni}^0, L^P), \qquad R_n^G = \text{updateRes_pub}(\lambda_{ni}^W, L_n^G).$$

  Compute local income:

$$Y_n^{loc} = \text{updateLocInc}(R_n^P, R_n^G, v_n^P, v_n^G).$$

  Compute private non-tradable and tradable employment:

$$L_{nN}^P = \text{updatePrivNEmpl}(Y_n^{loc}, w_n^0), \qquad L_{nT}^P = L_n^P - L_{nN}^P.$$

  Compute rents:

$$Q_n = \text{updateRent}(Y_n^{loc}, \bar{H}_n).$$

  Compute productivity:

$$A_n = \bar{A}_n \left(L_n^P + L_n^G \right)^{\nu}.$$

  Algorithm continues on the next page.

Algorithm 5: Counterfactual equilibrium with public employment (continued)

Compute trade shares:

$$\pi_{ni} = \text{updatetradesh}(A_n, L_{nT}^P, w_n^0, d_{ni}^T).$$

Compute tradable and non-tradable prices:

$$P_{nT} = \text{updatePrice}(L_{nT}^P, w_n^0, \pi_{ni}, d_{ni}^T, A_n), \qquad P_{nN} = \text{updatePrice_NT}(w_n^0, A_n).$$

Update wages:

$$w_n^1 = \text{updateWage}(\pi_{ni}, Y_n^{loc}, L_n^P).$$

Update commuting probabilities:

$$\lambda_{ni}^1 = \text{updateLambda_alt}(P_{nT}, (P_{nN}, Q_n, w_n^0, B_{ni}, d_{ni}).$$

if *any updated object is non-finite, non-real, or negative* **then**
    Store failure iteration, variable name, row, column, and value; then stop

if $\max_n |w_n^0 - w_n^1| < 10^{-4}$ *and* $\max_{n,i} |\lambda_{ni}^0 - \lambda_{ni}^1| < 10^{-4}$ **then**
    Set convergence status to true and stop

if $m \geq \bar{m}$ **then**
    Stop and report non-convergence

Apply damped updating:

$$w_n^0 = 0.25 w_n^1 + 0.75 w_n^0, \qquad \lambda_{ni}^0 = 0.25 \lambda_{ni}^1 + 0.75 \lambda_{ni}^0.$$

Result: Private sector wages w_n^0, private-sector commuting probabilities λ_{ni}^0, expected private-sector residential wages v_n^P, private employment L_n^P, tax rate t, tradable and non-tradable employment L_{nT}^P and L_{nN}^P, private and public residents R_n^P and R_n^G, rents Q_n, trade shares π_{ni}, tradable and non-tradable prices P_{nT} and P_{nN}, productivity A_n, convergence indicator, and failure diagnostics.

Algorithm 6: Prerun adjustment of commuting probabilities

Data: Same as Algorithm 5 and definition of needed successful rounds K

Initialize: Set adjustment size $\Delta = 0.001$, maximum prerun iterations $\bar{p}$, and success counter $c = 0$.

Purpose: Before starting the main counterfactual loop, adjust the initial commuting probability matrix so that the implied private tradable employment is nonnegative in every location for K consecutive update rounds. Potentially needed to balance out increased service demand of public-sector worker increase. Code is nested within `LamW_CF_full_V2.m`. Code does not bite in any of the calculations of the paper, tradable employment remains positive after initial relocations in both counterfactuals.

Check that λ^0_{ni} is square, finite, real, and nonnegative

for $p = 1, \ldots, \bar{p}$ **do**

  Run the counterfactual update system of Algorithm 5 up to temporary tradable employment $L_n^{P,T,tmp}$

  if *any temporary object is non-finite, non-real, or negative* **then**

    Store failure diagnostics and stop

  Identify locations with negative tradable employment:

$$\mathcal{N} = \{n : L_{nT}^{P,tmp} < 0\}.$$

  if $\mathcal{N} \neq \emptyset$ **then**

    Set success counter $c = 0$

    foreach $i \in \mathcal{N}$ **do**

      Let total commuting mass be

$$M = \sum_n \sum_j \lambda^0_{nj},$$

      row-i mass be

$$M_i = \sum_j \lambda^0_{ij},$$

      and remaining mass be

$$M_{-i} = M - M_i.$$

      Set feasible shift:

$$d = \min\{\Delta, 0.999 M_{-i}\}.$$

      Increase row i proportionally:

$$\lambda^0_{ij} \leftarrow \lambda^0_{ij} \left(1 + \frac{d}{M_i}\right) \qquad \text{if } M_i > 0.$$

      If $M_i = 0$, place the added mass on the diagonal (Fallback: no application):

$$\lambda^0_{ii} \leftarrow \lambda^0_{ii} + d.$$

      Algorithm continues on the next page.

Algorithm 6: Prerun adjustment of commuting probabilities (continued)

Remove the same mass proportionally from all other rows:

$$\lambda^0_{nj} \leftarrow \lambda^0_{nj}\left(1 - \frac{d}{M_{-i}}\right), \qquad n \neq i.$$

if $\mathcal{N} \neq \emptyset$ **then**
    Check that the adjusted matrix is finite and nonnegative
    Continue to next prerun iteration
Complete the temporary counterfactual update round and compute temporary rents, productivity, trade shares, prices, wages, and commuting probabilities
Increase success counter:

$$c \leftarrow c + 1.$$

Advance the state using damped updating:

$$w^0_n \leftarrow 0.25 w^{1,tmp}_n + 0.75 w^0_n, \qquad \lambda^0_{ni} \leftarrow 0.25 \lambda^{1,tmp}_{ni} + 0.75 \lambda^0_{ni}.$$

if $c \geq K$ **then**
    Return success

If fewer than K consecutive successful rounds are reached, return failure

Result: Adjusted wage guesses w^0_n, adjusted commuting probabilities λ^0_{ni}, and a success/failure indicator.

Algorithm 7: Computing conditional workplace probabilities: `updateCondCom.m`

Data: Unconditional commuting probability matrix λ^0_{ni}.

Compute total commuting probability into each workplace location:

$$s_i = \sum_n \lambda^0_{ni}.$$

Normalize each column by its workplace total:

$$\lambda^W_{ni} = \frac{\lambda^0_{ni}}{s_i} = \frac{\lambda^0_{ni}}{\sum_n \lambda^0_{ni}}.$$

Result: Conditional workplace probability matrix λ^W_{ni}.

Algorithm 8: Computing workplace employment: `updateEmpl.m`

Data: Unconditional commuting probability matrix λ_{ni}^0 and total private employment $\bar{L}$.

94 Compute the total probability mass assigned to each workplace location:

$$s_i = \sum_n \lambda_{ni}^0.$$

95 Scale workplace probability masses by total employment:

$$L_i = \bar{L} s_i = \bar{L} \sum_n \lambda_{ni}^0.$$

Result: Workplace employment vector L_i.

Algorithm 9: Computing the counterfactual tax rate: `updateTax.m`

Data: Public employment L_n^G, public wages w_n^G, private wages w_n^0, and private workplace employment L_n^P.

96 Compute local public wage bills:

$$G_n = w_n^G L_n^G.$$

97 Compute the total public budget:

$$G = \sum_n G_n = \sum_n w_n^G L_n^G.$$

98 Compute the aggregate wage tax base:

$$\mathcal{T} = \sum_n w_n^0 L_n^P + \sum_n w_n^G L_n^G.$$

99 Compute the uniform tax rate:

$$t = \frac{G}{\mathcal{T}}.$$

Result: Uniform tax rate t.

Algorithm 10: Computing expected private residential wages: `updateResWage_priv_alt.m`

Data: Private wage guesses w_i^0, bilateral commuting travel time d_{ni}, uniform tax rate t, bilateral amenities B_{ni}, and structural parameters $\{\epsilon, \gamma\}$.

100 Compute bilateral commuting attractiveness:

$$a_{ni} = B_{ni}(w_i^0)^{\epsilon} d_{ni}^{-\epsilon\gamma}.$$

101 Compute the row-wise denominator for each residence location:

$$D_n = \sum_i a_{ni}.$$

102 Compute conditional commuting probabilities given residence:

$$\lambda_{ni|n} = \frac{a_{ni}}{D_n} = \frac{B_{ni}(w_i^0)^{\epsilon} d_{ni}^{-\epsilon\gamma}}{\sum_s B_{ns}(w_s^0)^{\epsilon} d_{ns}^{-\epsilon\gamma}}.$$

103 Compute expected after-tax private wage at residence:

$$v_n^P = \sum_i \lambda_{ni|n}(1-t)w_i^0.$$

Result: Expected private residential wage v_n^P.

Algorithm 11: Computing after-tax public residential wages: `updateResWage_pub.m`

Data: Public-sector wages w_n^G and uniform tax rate t.

104 Compute after-tax public residential wages:

$$v_n^G = (1-t)w_n^G.$$

Result: After-tax public residential wage vector v_n^G.

Algorithm 12: Computing private residents: `updateRes_priv.m`

Data: Unconditional commuting probability matrix λ_{ni}^0 and total private employment $\bar{L}$.

105 Compute the total probability mass assigned to each residence location. Scale residence probability masses by total private employment:

$$R_n^P = \bar{L} \sum_i \lambda_{ni}^0.$$

Result: Private residents R_n^P.

Algorithm 13: Computing public residents: `updateRes_pub.m`

Data: Conditional residence probabilities by workplace λ_{ni}^W and public employment L_i^G.

106 Allocate public workers to residence locations using the workplace-conditional commuting probabilities:

$$R_n^G = \sum_i \lambda_{ni}^W L_i^G.$$

Result: Public residents R_n^G.

Algorithm 14: Computing local income: `updateLocInc.m`

Data: Private residents R_n^P, public residents R_n^G, after-tax private residential wages v_n^P, and public residential wages v_n^G.

107 Compute local income as the sum of private and public residents' after-tax income:

$$Y_n^{loc} = v_n^P R_n^P + v_n^G R_n^G.$$

Result: Local income Y_n^{loc}.

Algorithm 15: Computing private non-tradable employment: `updatePrivNEmpl.m`

Data: Local income Y_n^{loc}, private wage guesses w_n^0, and expenditure parameters $\{\alpha_N, \alpha_T\}$.

108 Compute local expenditure on non-tradables:

$$X_n^N = \frac{\alpha_N}{\alpha_N + \alpha_T} Y_n^{loc}.$$

109 Convert non-tradable expenditure into private non-tradable employment:

$$L_{nN}^P = \frac{X_n^N}{w_n^0} = \frac{\frac{\alpha_N}{\alpha_N+\alpha_T} Y_n^{loc}}{w_n^0}.$$

Result: Private non-tradable employment L_{nN}^P.

Algorithm 16: Computing local rents: `updateRent.m`

Data: Local income Y_n^{loc}, land supply $\bar{H}_n$, and structural parameters $\{\alpha_T, \alpha_N, \delta\}$.

110 Compute local housing expenditure:

$$X_n^H = (1 - \alpha_T - \alpha_N) Y_n^{loc}.$$

111 Compute local rents from the inverse housing supply equation:

$$Q_n = \left(\frac{X_n^H}{\bar{H}_n} \right)^{\frac{1}{1+\delta}} = \left(\frac{(1 - \alpha_T - \alpha_N) Y_n^{loc}}{\bar{H}_n} \right)^{\frac{1}{1+\delta}}.$$

Result: Local rents Q_n.

Algorithm 17: Computing bilateral trade shares: `updatetradesh.m`

Data: Productivity A_i, private tradable employment L_{iT}^P, private wage guesses w_i^0, bilateral freight travel time d_{ni}^T, and structural parameters $\{\sigma, \psi\}$.

112 Compute producer-specific trade attractiveness:

$$a_i = A_i^{\sigma-1} L_{iT}^P (w_i^0)^{1-\sigma}.$$

113 Construct the bilateral trade attractiveness matrix:

$$q_n = {d_{ni}^T}^{\psi(1-\sigma)} a_i.$$

114 Compute total attractiveness for each spending location:

$$D_n = \sum_i q_{ni}.$$

115 Normalize by destination-specific total attractiveness:

$$\pi_{ni} = \frac{q_{ni}}{D_n} = \frac{{d_{ni}^T}^{\psi(1-\sigma)} A_i^{\sigma-1} L_{iT}^P (w_i^0)^{1-\sigma}}{\sum_k {d_{nk}^T}^{\psi(1-\sigma)} A_k^{\sigma-1} L_{kT}^P (w_k^0)^{1-\sigma}}.$$

Result: Bilateral trade share matrix π_{ni}.

Algorithm 18: Computing the tradable price index: `updatePrice.m`

Data: Private tradable employment L_{nT}^P, private wage guesses w_n^0, bilateral trade shares π_{ni}, bilateral freight travel time d_{ni}^T, productivities A_n, and structural parameters $\{\sigma, F, \psi\}$.

116 Extract own trade shares:

$$\pi_{nn} = \text{diag}(\pi_{ni}).$$

117 Extract internal freight travel time:

$$d_{nn}^T = \text{diag}(d_{ni}^T).$$

118 Compute the tradable price index:

$$P_{nT} = \frac{\sigma}{\sigma - 1} \left(\frac{L_{nT}^P}{\sigma F \pi_{nn}} \right)^{\frac{1}{1-\sigma}} \frac{{d_{nn}^T}^{\psi} w_n^0}{A_n}.$$

Result: Tradable price index P_n^T.

Algorithm 19: Computing the non-tradable price index: `updatePrice_NT.m`

Data: Private wage guesses w_n^0 and productivities A_n.

119 Compute the non-tradable price index:

$$P_n^N = \frac{w_n^0}{A_n}.$$

Result: Non-tradable price index P_{nN}.

Algorithm 20: Updating private wages: `updateWage.m`

Data: Bilateral trade shares π_{ni}, local income Y_n^{loc}, private workplace employment L_i^P, and expenditure parameters $\{\alpha_T, \alpha_N\}$.

120 Compute tradable expenditure in each spending location:

$$X_n^T = \frac{\alpha_T}{\alpha_T + \alpha_N} Y_n^{loc}.$$

121 Compute tradable expenditure received by each producing location:

$$Rev_i^T = \sum_n \pi_{ni} X_n^T.$$

122 Compute local non-tradable expenditure:

$$X_i^N = \frac{\alpha_N}{\alpha_T + \alpha_N} Y_i^{loc}.$$

123 Compute total private revenue by workplace location:

$$Rev_i = Rev_i^T + X_i^N.$$

124 Update private wages:

$$w_i^1 = \frac{Rev_i}{L_i^P} = \frac{\sum_n \pi_{ni} \frac{\alpha_T}{\alpha_T + \alpha_N} Y_n^{loc} + \frac{\alpha_N}{\alpha_T + \alpha_N} Y_i^{loc}}{L_i^P}.$$

Result: Updated private wage vector w_i^1.

Algorithm 21: Updating unconditional commuting probabilities: `updateLambda_alt.m`

Data: Tradable price index P_n^T, non-tradable price index P_n^N, rents Q_n, private wage guesses w_i^0, bilateral amenities B_{ni}, bilateral commuting costs d_{ni}, and structural parameters $\{\alpha_T, \alpha_N, \epsilon, \gamma\}$.

125 Compute the residence-specific cost-of-living component:

$$C_n = (P_n^T)^{\alpha_T}(P_n^N)^{\alpha_N} Q_n^{1-\alpha_T-\alpha_N}.$$

126 Replicate C_n across workplace destinations and w_i^0 across residence origins.

127 Compute bilateral location-choice attractiveness:

$$a_{ni} = B_{ni} d_{ni}^{-\epsilon\gamma} C_n^{-\epsilon} (w_i^1)^{\epsilon}.$$

128 Equivalently:

$$a_{ni} = B_{ni} d_{ni}^{-\epsilon\gamma} \left[(P_n^T)^{\alpha_T}(P_n^N)^{\alpha_N} Q_n^{1-\alpha_T-\alpha_N}\right]^{-\epsilon} (w_i^1)^{\epsilon}.$$

129 Compute total attractiveness:

$$A = \sum_n \sum_i a_{ni}.$$

130 Normalize to obtain unconditional commuting probabilities:

$$\lambda_{ni}^1 = \frac{a_{ni}}{A} = \frac{B_{ni} d_{ni}^{-\epsilon\gamma} \left[(P_n^T)^{\alpha_T}(P_n^N)^{\alpha_N} Q_n^{1-\alpha_T-\alpha_N}\right]^{-\epsilon} (w_i^1)^{\epsilon}}{\sum_r \sum_s B_{rs} d_{rs}^{-\epsilon\gamma} \left[(P_r^T)^{\alpha_T}(P_r^N)^{\alpha_N} Q_r^{1-\alpha_T-\alpha_N}\right]^{-\epsilon} (w_s^1)^{\epsilon}}.$$

Result: Updated unconditional commuting probability matrix λ_{ni}^1.

Algorithm 22: Computing aggregate expected utility: `Utilcalc.m`

Data: Bilateral amenities B_{rs}, bilateral commuting travel time d_{rs}, tradable price index P_r^T, non-tradable price index P_r^N, private wages w_s^P, rents Q_r, uniform tax rate t, and structural parameters $\{\alpha_T, \alpha_N, \epsilon, \gamma\}$.

131 Compute the residence-side cost-of-living component:

$$C_r = (P_r^T)^{\alpha_T}(P_r^N)^{\alpha_N} Q_r^{1-\alpha_T-\alpha_N}.$$

132 Compute bilateral commuting and cost-of-living disutility:

$$D_{rs} = \left(d_{rs}^{\gamma} C_r\right)^{-\epsilon}.$$

133 Compute after-tax workplace wage attractiveness:

$$W_s = \left[(1-t)w_s^P\right]^{\epsilon}.$$

134 Compute the bilateral utility summand:

$$S_{rs} = B_{rs} D_{rs} W_s.$$

135 Compute total attractiveness:

$$S = \sum_r \sum_s S_{rs} = \sum_r \sum_s B_{rs} \left[d_{rs}^{\gamma}(P_r^T)^{\alpha_T}(P_r^N)^{\alpha_N} Q_r^{1-\alpha_T-\alpha_N}\right]^{-\epsilon} \left[(1-t)w_s^P\right]^{\epsilon}.$$

136 Compute aggregate expected utility:

$$\bar{U} = \Gamma\left(\frac{\epsilon-1}{\epsilon}\right) S^{1/\epsilon}.$$

Result: Aggregate expected utility $\bar{U}$.

B District Harmonization

South Korean administrative units comprise three layers. The top layer consists of provinces (도, *do*) and municipalities (시, *si*). Provinces are divided into rural counties (군, *gun*) and cities (시, *si*), which may be further divided, depending on size, into urban boroughs (구, *gu*). Municipalities are divided into either rural counties (군, *gun*) or urban boroughs (구, *gu*). At the third layer, urban boroughs are subdivided into neighborhoods (동, *dong*), while rural counties consist of towns (면, *myeon*) and townships (읍, *eup*).

All data sources are aggregated to the level of the most coarsely aggregated dataset, the Regional Wage Survey, which reports only *si* and *gun* units. Municipalities therefore each comprise a single unit, while provinces consist of rural areas and cities, yielding a baseline of 168 units in 2010.

Several further adjustments are applied to the baseline of 168 units. The island county of Ulleunggun, located far to the east, is excluded on account of its remoteness and population of fewer than 10,000 inhabitants. The two administrative units on Jeju Island are merged into one: travel data for air and sea routes are unavailable, precluding any distinction between the two units' connectivity to the mainland. A set of harmonizations then merges all units whose borders change between 2010 and 2019, facilitating comparisons over time with the observed outcomes of the Innovation City Project. Osan is merged into Hwasong and Guri into Namyangju; each pair was originally a single administrative unit that split prior to the harmonization period. The merge is motivated by the model's treatment of non-tradable consumption as occurring exclusively at the residence location: in each pair, a larger county surrounds a smaller urban city, allowing a significant share of residents whose residences lie near the border to consume non-tradable goods in the neighboring unit. Gwangmyeong is merged into Seoul on the same grounds. Finally, Gwacheon is integrated into Seoul because Gwacheon houses the administrative complex from which several relocated institutions departed; since the origin district of each relocated agency cannot be tracked individually, Gwacheon and Seoul are treated as a single unit and all associated relocated jobs are attributed to Seoul. With these adjustments applied, the final count of spatial units reduces to 154.

C Travel Time Calculation

Construction of the travel time matrix begins by translating the KORAIL rail schedules into travel times between stations. All direct connections between station pairs across all rail lines are collected, and the fastest connection for each pair is retained; all travel times are recorded in minutes. The data span 2004–2015; although only 2010 and 2011 travel times are required, rail lines from earlier years that were subsequently converted to subway lines are also included. Subway lines are administered by municipal rail authorities and their schedules are therefore absent from the KORAIL data. Although most subway lines remain absent from the data, this approach captures at least the longest converted lines, which tend to exceed the length of subway lines built directly by municipal authorities. With minimum inter-station travel times for 2010 established, stations in the KORAIL schedules are matched by name

to stations in shapefiles obtained from the Korea Transport Institute. Where multiple or no matches exist, the correct station is identified and matched manually; where a station is absent from the shapefile, it is matched to the nearest available station.

The road network data from the Korea Transport Institute comprise geocoded nodes and edges representing the South Korean road network. Edges represent individual road segments; nodes represent either traffic-relevant locations, such as crossings, splits, or endpoints, or map-specific reference points, such as grid-cell or district borders. Travel time along each edge is calculated from the segment length and the maximum permitted speed on that segment.

With road travel times established, each harmonized county centroid and each station is connected to the five closest points in the combined network, excluding map-specific road nodes, at an assigned travel speed of 20 km/h. This speed corresponds to a restrictive inner-city travel pace and serves as a reasonable approximation for travel on water, simplifying the connection of Jeju to the mainland without requiring separate treatment of sea routes.

With all district centroids and stations connected to the road network, the `sfnetworks` package in R is used to calculate travel times between all counties. This calculation is performed twice: once for commuting times and once for freight transport, with each pass adjusting for the modes of transportation available. The commuting calculation follows the approach described above. For freight transport, the minimum inter-station travel time calculation is repeated after excluding all high-speed rail lines, which in 2010 comprise only the Gyeongbu line connecting Seoul to Busan. Road speeds are also capped for freight: segments with a maximum speed exceeding 100 km/h are limited to 100 km/h, and segments with speeds between $80 - 100$ km/h are limited to 80 km/h, following the traffic regulations for trucks in South Korea (USFK 2021).

Internal travel times for each district are calculated as 20 % of the average travel time to all bordering districts, following Seidel and Wickerath (2020). This approach approximates the travel time from the district center to its border and serves as a reliable proxy for internal commuting times. To accommodate coastal island districts whose borders may not adjoin the mainland, a buffer of 500 meters is applied to capture bridge connections. Internal travel times can thus be computed for all districts except Jeju, which comprises a single district with no neighbors; its internal travel time is accordingly set to 8 minutes, the average of the non-municipal districts.

D Additional Place-Based Policies

Aside from the Innovation City Project, the South Korean government continuously implemented a range of policies aimed at curbing concentration in the Seoul Metropolitan Area and bolstering growth elsewhere in the country. The most closely related is the construction of Sejong (Bang 2025). Initially devised as a new capital in 2004, the proposal to remove capital status from Seoul was ruled unconstitutional by the Constitutional Court in the same year. Following the Constitutional Court ruling, the planned role for Sejong was amended, designating it a multifunctional

administrative city in which Seoul would retain capital status along with the most important central government institutions, including the President's residence, the National Assembly, the Constitutional Court, and security-related ministries. The remaining ministries were to be moved to Sejong, together with the share of public agencies allocated to Sejong from those deemed suitable for relocation out of Seoul. Initial construction began in 2007, with the "Special Act on the Establishment" of Sejong City enacted in 2010, establishing the city's local government authority. Contrary to the Innovation City Project, Sejong was built as an entirely new planned city, independent of any existing urban agglomeration; designed for a resident capacity of 500,000, its population reached 338,000 by 2019.

The third policy within the broader New Town programme was the Enterprise City Project (Lee, You, and Kwon 2015), which aimed to develop purpose-built urban districts specializing in industry, science, or tourism. Unlike Sejong and the Innovation Cities, Enterprise Cities were privately funded, with the state providing only the legal framework: designating development zones and reducing tax burdens for developers and local industry. By 2008, seven Enterprise Cities had been designated and were under construction (Song et al. 2020).

Alongside the New Town programme, a parallel set of laws was introduced to manage growth in the Seoul Metropolitan Area and limit over-concentration. The first such laws date to the 1980s. The primary instrument is the "Capital Region Readjustment Planning Act", introduced in 1982 and continuously amended since, with restrictions increasing in intensity over time. The Act divides the Seoul Metropolitan Area into zones of varying regulatory stringency. In 2011, these comprised the Overconcentration Control Region, the Nature Preservation Region, and Growth Management Areas. The Overconcentration Control Region, which encompasses the entire city of Seoul, and the Nature Preservation Region both prohibit the construction of any "population-intensive facilities", a category that includes factories or universities above a certain size, unless deemed necessary for national growth or public welfare. Growth Management Areas apply less stringent regulations but permit construction only within specified growth targets. Beyond outright restrictions on construction, an overcrowding surcharge is levied on any entity constructing new facilities in the Seoul Metropolitan Area, set at 10 % of construction cost with limited exceptions. Large-scale development is subject to approval by the Ministry of Land, Infrastructure and Transport, giving the central government oversight of development in Seoul. The Act also establishes total capacity regulations defining the maximum land available for each development type in the capital, including the maximum area available for industrial purposes.

Beyond the Capital Region Readjustment Planning Act, several other laws address the unequal growth between the Seoul Metropolitan Area and the rest of the country, particularly in the tax code. Two are especially relevant: the "Local Tax Act" and the "Special Tax Exceptions and Restrictions Act". The Local Tax Act levies higher acquisition and property taxes on new factories and businesses in the Seoul Metropolitan Area, as well as higher corporate registration taxes. The Act further differentiates tax treatment for industrial complexes: property taxes are reduced by 50 % for five years in complexes located within the Seoul Metropolitan Area, while new establishments in complexes outside the Seoul Metropolitan Area

are fully exempt from property taxes for the same period. The Special Tax Exceptions and Restrictions Act reduces corporate or income taxes for new small and medium enterprises outside the Seoul Metropolitan Area and creates tax incentives for businesses and corporations relocating headquarters or establishments out of the Seoul Metropolitan Area.

Several large infrastructure projects coincided with the Innovation City Project. The South Korean rail network saw major expansions, most notably the completion of the Gyeongbu high-speed rail line in November 2010, which added the final segment connecting Ulsan and Busan. The second major upgrade was the completion of the Honam high-speed rail line in 2015, linking the southwest to the capital. The Lee Myung-bak government, which came to power in 2008, oversaw a substantial expansion of the highway network under its 5+2 Economic Regions Programme (Park 2024), constructing outer beltway roads to ease traffic around metropolitan centers and connecting roads linking urban and rural regions, ports, and airports.

The construction of industrial complexes represents another large and ongoing infrastructure programme: over 400 were designated in the period 2010–2019, bringing the total number to over 1,000. These complexes provide financial and organizational support to resident manufacturing firms, as well as access to shared infrastructure such as power generation and road and rail connections. Initially deployed as a tool for guiding industrial policy in the 1960s, with designations concentrated in the Seoul Metropolitan Area and the Dongnam region, the programme's focus shifted over time. From the 1980s onward, new complexes were located farther from urban centers to alleviate air pollution and congestion. By 2010, balanced growth had been written explicitly into the laws governing industrial complex construction and management. This reorientation mirrored the logic of the tax legislation, combining stronger incentives for complexes outside the Seoul Metropolitan Area with tighter restrictions on those within it.

E Additional Figures and Tables

Table E.1: Total Population and Capital-Area Population of Select OECD Countries

Country	Capital FUA	Population	Capital FUA population	Share (%)
South Korea	Seoul	49.55	23.04	46.5
Denmark	Copenhagen	5.54	1.93	34.8
Japan	Tokyo	128.06	35.42	27.7
France	Paris	64.77	12.61	19.5
United Kingdom	London	62.76	11.26	17.9
Mexico	Mexico City	114.76	19.06	16.6
Spain	Madrid	46.56	6.65	14.3
Italy	Rome	59.82	4.08	6.8
Germany	Berlin	81.78	4.64	5.7

Notes: Table shows total population, capital functional urban area population, and the capital functional urban area's share in total population for select OECD countries, including South Korea. Functional urban areas describe continuous urban agglomerations as defined by the OECD. Source: OECD Data Explorer .Functional Urban Area population data for 2010. Population numbers are in millions.

Table E.2: Seoul Metropolitan Area Concentration of Activity

	Seoul Metropolitan Area	Rest of Korea
Top 100 company HQs	91 %	9 %
Venture firms / startups	70 %	30 %
Manufacturing firms	57 %	43 %
Financial loans/lending	67 %	33 %
Public institutions	85 %	15 %
Top 20 universities	80 %	20 %
Airport arrivals (foreigners)	87 %	13 %

Notes: The table provides the distribution of select economic, educational, political and cultural markers between the Seoul Metropolitan Area and the rest of South Korea. Source: Top universities ranked by JoongAng Ilbo newspaper. Airport arrivals from Korea Tourism Statistics. Remaining statistics from Kim et al. (2006). University rankings and airport arrivals as of 2010, remaining statistics as of 2005.

Table E.3: Innovation City Location Selection Criteria

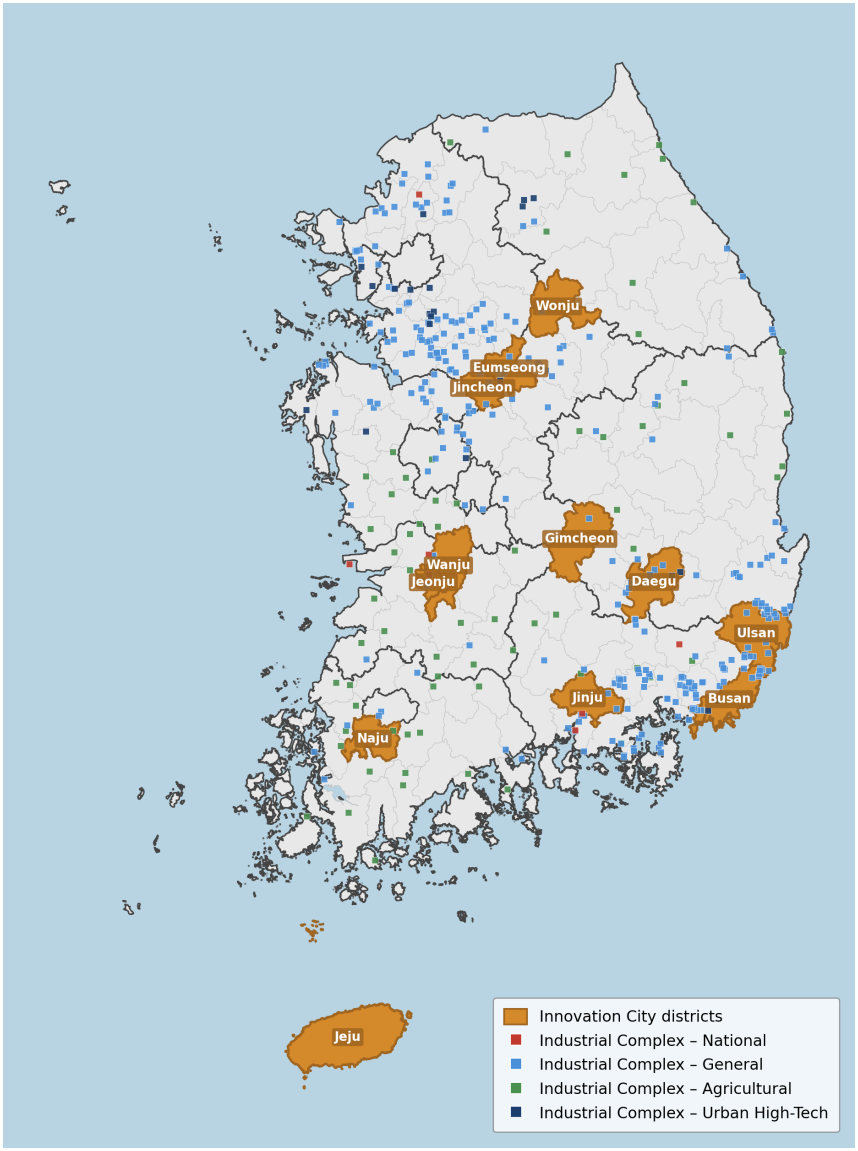
Category	Criterion	Points
Potential to develop into an innovation hub	Accessibility to major transport networks	20
	Suitability as an innovation hub	20
	Ability to use existing urban infrastructure and amenities	10
Appropriateness of urban development	Ease and economic feasibility of urban development	15
	Environmentally friendly locational suitability	10
Potential for shared growth within region	Balanced development within the region	10
	Plan for sharing Innovation City benefits	10
	Support from local governments	5
	Total	**100**

Notes: The table shows the selection criteria and their weight in the selection process of Innovation City counties within provinces. Source: PAPPC (2008)

Figure E.1: Innovation City Construction Efforts

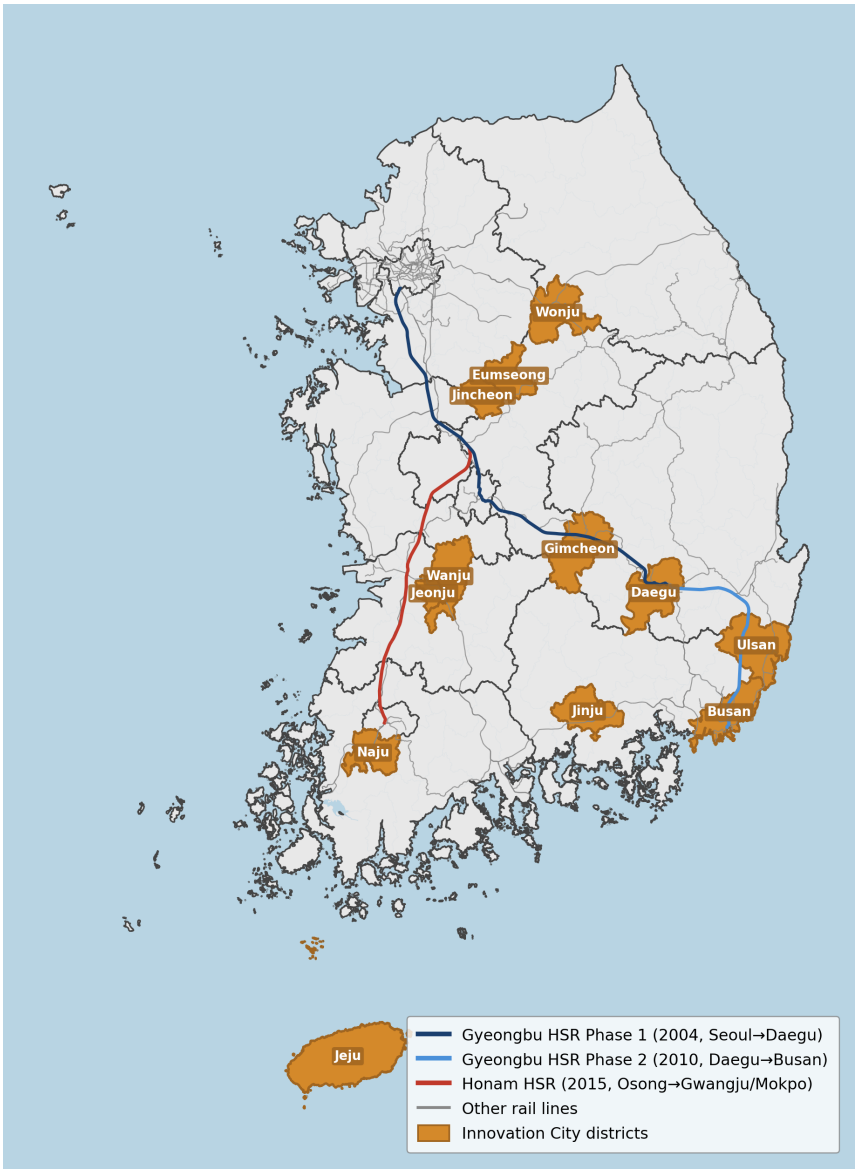
Notes: Panel (a) shows Innovation City Wonju in the Province of Gangwon prior to the construction start in 2008. Panel (b) shows the extent of the Innovation City as of 2016. Panel (c) shows the status pre-construction for the Innovation city in Namgu, Busan. Panel (d) shows the extent of the Innovation City as of 2017. Source: Kakao Maps

Figure E.2: Industrial Complex Designation



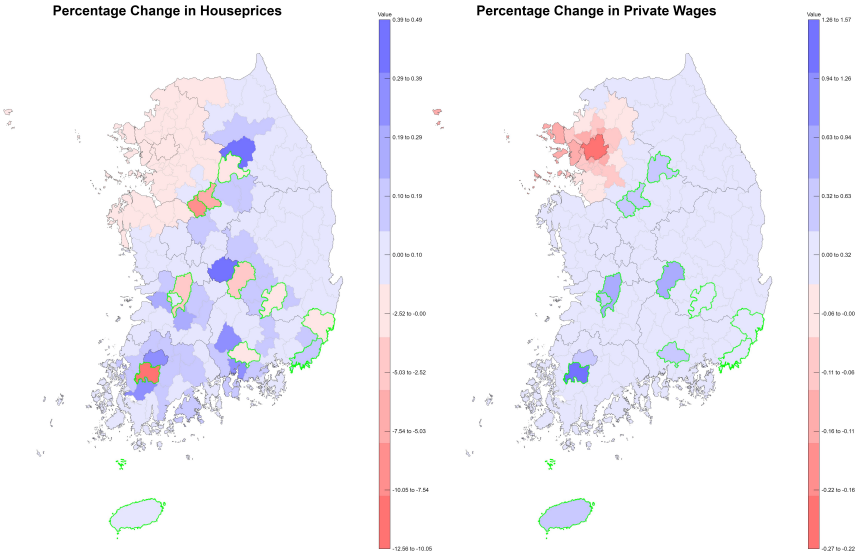
Notes: Map shows designations of industrial complexes by type between 2010 and 2019. Innovation Cities highlighted in orange. Industrial complexes differentiated by type: National (red), General (light blue), Agricultural (green) and Urban High-Tech (dark blue). Source: Industrial Land Information System

Figure E.3: South Korean High-Speed Rail Network



Notes: Map shows South Korean high-speed rail network as of 2018. Innovation Cities highlighted in orange. Phases of high-speed rail expansion in blue, light blue and red. Conventional rail lines in gray. Source: Korea Transport Institute

Figure E.4: House-Price and Wage Changes of Counterfactual Innovation City Project



Notes: Percentage changes in county level house-prices and private-sector wages post counterfactual Innovation City Project, including public-sector relocation and housing construction. Targeted Innovation Cities highlighted in green. Increases in blue, with decreases in red. Left house-prices, right private wages.